

Trigonometric Ratios & Identities

Question1

$$\sin \frac{\pi}{12} \sin \frac{2\pi}{12} \sin \frac{3\pi}{12} \sin \frac{4\pi}{12} \sin \frac{5\pi}{12} \sin \frac{6\pi}{12} =$$

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Options:

A.

$$\frac{\sqrt{3}}{16\sqrt{2}}$$

B.

$$\frac{\sqrt{3}}{8\sqrt{2}}$$

C.

$$\frac{1}{32}$$

D.

$$\frac{1}{16}$$

Answer: A

Solution:

$$\begin{aligned} & \sin \frac{\pi}{12} \cdot \sin \frac{2\pi}{12} \cdot \sin \frac{3\pi}{12} \cdot \sin \frac{4\pi}{12} \cdot \sin \frac{5\pi}{12} \cdot \sin \frac{6\pi}{12} \\ &= \sin 15^\circ \cdot \sin 30^\circ \cdot \sin 45^\circ \cdot \sin 60^\circ \sin 75^\circ \cdot \sin 90^\circ \\ &= \frac{\sqrt{6} - \sqrt{2}}{4} \times \frac{1}{2} \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} \times \frac{\sqrt{6} + \sqrt{2}}{4} \times 1 \\ &= \frac{(\sqrt{6})^2 - (\sqrt{2})^2}{4 \times 4} \times \frac{\sqrt{3}}{4\sqrt{2}} \\ &= \frac{1}{4} \times \frac{\sqrt{3}}{4\sqrt{2}} = \frac{\sqrt{3}}{16\sqrt{2}} \end{aligned}$$

Question2

If $A + B + C + D = 2\pi$, then $\sin A + \sin B + \sin C + \sin D =$

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Options:

A.

$$4 \sin \left(\frac{A+B}{4} \right) \sin \left(\frac{A+C}{4} \right) \sin \left(\frac{A+D}{4} \right)$$

B.

$$4 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A+C}{4} \right) \cos \left(\frac{A+D}{4} \right)$$

C.

$$4 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A+C}{2} \right) \sin \left(\frac{A+D}{2} \right)$$

D.

$$4 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A+C}{4} \right) \sin \left(\frac{A+D}{4} \right)$$

Answer: C

Solution:

$$\begin{aligned} & (\sin A + \sin D) + (\sin B + \sin C) \\ &= 2 \sin \left(\frac{A+D}{2} \right) \cos \left(\frac{A-D}{2} \right) + 2 \sin \left(\frac{B+C}{2} \right) \cos \left(\frac{B-C}{2} \right) \end{aligned}$$

$$\begin{aligned} & \because A + B + C + D = 2\pi \\ & \Rightarrow B + C = 2\pi - (A + D) \\ &= 2 \sin \left(\frac{A+D}{2} \right) \cos \left(\frac{A-D}{2} \right) + 2 \sin \left(\pi - \frac{A+D}{2} \right) \cos \left(\frac{B-C}{2} \right) \end{aligned}$$

$$\begin{aligned} &= 2 \sin \left(\frac{A+D}{2} \right) \left[\cos \left(\frac{A-D}{2} \right) + \cos \left(\frac{B-C}{2} \right) \right] \\ &= 2 \sin \left(\frac{A+D}{2} \right) \left[2 \cos \left(\frac{\frac{A-D}{2} + \frac{B-C}{2}}{2} \right) \cos \left(\frac{\frac{A-D}{2} - \frac{B-C}{2}}{2} \right) \right] \\ &= 4 \sin \left(\frac{A+D}{2} \right) \left[\cos \left(\frac{A+B-(C+D)}{4} \right) \cos \left(\frac{A+C-(B+D)}{4} \right) \right] \end{aligned}$$

$$\begin{aligned} & \because C + D = 2\pi - (A + B) \\ & \text{and } B + D = 2\pi - (A + C) \\ &= 4 \sin \left(\frac{A+D}{2} \right) \left[\cos \left(\frac{\pi}{2} - \left(\frac{A+B}{2} \right) \right) \cos \left(\frac{\pi}{2} - \left(\frac{A+C}{2} \right) \right) \right] \\ &= 4 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A+C}{2} \right) \sin \left(\frac{A+D}{2} \right) \end{aligned}$$

Question3

$$\left(4 \cos^2 \frac{\pi}{20} - 1\right) \left(4 \cos^2 \frac{3\pi}{20} - 1\right) \left(4 \cos^2 \frac{5\pi}{20} + 1\right) \left(4 \cos^2 \frac{7\pi}{20} - 1\right) \left(4 \cos^2 \frac{9\pi}{20} - 1\right) =$$

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Options:

A.

1

B.

1/2

C.

2

D.

3

Answer: D

Solution:

We have,

$$\begin{aligned} & \left(4 \cos^2 \frac{\pi}{20} - 1\right) \left(4 \cos^2 \frac{3\pi}{20} - 1\right) \\ & \left(4 \cos^2 \frac{5\pi}{20} + 1\right) \left(4 \cos^2 \frac{7\pi}{20} - 1\right) \left(4 \cos^2 \frac{9\pi}{20} - 1\right) \\ &= \left(4 \cos^2 \frac{\pi}{20} - 1\right) \left(4 \cos^2 \frac{3\pi}{20} - 1\right) \left(4 \times \frac{1}{2} + 1\right) \\ & \left(4 \cos^2 \frac{7\pi}{20} - 1\right) \left(4 \cos^2 \frac{9\pi}{20} - 1\right) \\ &= 3 \left(4 \cos^2 \frac{\pi}{20} - 1\right) \left(4 \cos^2 \frac{9\pi}{20} - 1\right) \\ & \left(4 \cos^2 \frac{3\pi}{20} - 1\right) \left(4 \cos^2 \frac{7\pi}{20} - 1\right) \\ &= 3 \left(4 \cos^2 \frac{\pi}{20} - 1\right) \left(4 \sin^2 \frac{\pi}{20} - 1\right) \\ & \left(4 \cos^2 \frac{3\pi}{20} - 1\right) \left(4 \sin^2 \frac{3\pi}{20} - 1\right) \\ &= 3 \left(16 \cos^2 \frac{\pi}{20} \cdot \sin^2 \frac{\pi}{20} - 4 + 1\right) \left(16 \sin^2 \frac{3\pi}{20} \cos^2 \frac{3\pi}{20} - 4 - 1\right) \end{aligned}$$

$$\begin{aligned}
&= 3 \left(4 \sin^2 \frac{\pi}{10} - 3 \right) \left(4 \sin^2 \frac{3\pi}{10} - 3 \right) \\
&= \frac{3}{\sin \frac{\pi}{10} \cdot \sin \frac{3\pi}{10}} \left(4 \sin^3 \frac{\pi}{10} - 3 \sin \frac{\pi}{10} \right) \left(4 \sin^3 \frac{3\pi}{10} - 3 \sin \frac{3\pi}{10} \right) \\
&= \frac{3}{\sin \frac{\pi}{10} \sin \frac{3\pi}{10}} \cdot \sin \frac{3\pi}{10} \cdot \sin \left(\frac{9\pi}{10} \right) \\
&= 3 \left(\because \sin \left(\frac{9\pi}{10} \right) = \sin \left(\pi - \frac{\pi}{10} \right) = \sin \frac{\pi}{10} \right)
\end{aligned}$$

Question4

If A and B are the values such that $(A + B)$ and $(A - B)$ are not odd multiples of $\frac{\pi}{2}$ and $2 \tan(A + B) = 3 \tan(A - B)$, then $\sin A \cos A =$

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Options:

A.

$$\sin B \cos B$$

B.

$$5 \sin B \cos B$$

C.

$$\sin 2B$$

D.

$$\cos 2B$$

Answer: B

Solution:

Given,

$$2 \tan(A + B) = 3 \tan(A - B)$$

$$\Rightarrow \frac{\tan(A + B)}{\tan(A - B)} = \frac{3}{2}$$

$$\Rightarrow \frac{\tan(A + B) + \tan(A - B)}{\tan(A + B) - \tan(A - B)} = \frac{3 + 2}{3 - 2}$$

$$\Rightarrow \frac{[\sin(A + B) \cos(A - B)]}{[\sin(A + B) \cos(A - B)]} - \sin(A - B) \cos(A + B)]$$

$$\Rightarrow \frac{\sin(A + B + A - B)}{\sin(A + B - A + B)} = 5$$

$$\Rightarrow \sin 2A = 5 \sin 2B$$

$$\Rightarrow \sin A \cos A = 5 \sin B \cos B$$

Question5

If $\cos^3 80^\circ + \cos^3 40^\circ - \cos^3 20^\circ = k$, then $\frac{4k}{3} =$

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Options:

A.

$$\sin\left(\frac{4\pi}{3}\right)$$

B.

$$\cos\left(\frac{2\pi}{3}\right)$$

C.

$$\tan\left(\frac{\pi}{3}\right)$$

D.

$$\sec\left(\frac{2\pi}{3}\right)$$

Answer: B

Solution:

We have,

$$\cos^3 80^\circ + \cos^3 40^\circ - \cos^3 20^\circ = k$$

$$(\cos 80^\circ + \cos 40^\circ) \left((\cos 80^\circ + \cos 40^\circ)^2 - 3 \cos 80^\circ \cos 40^\circ \right) - \cos^3 20^\circ = k$$

$$\Rightarrow 2 \cos 60^\circ \cos 20^\circ \left((2 \cos 60^\circ \cos 20^\circ)^2 - \frac{3}{2} (\cos 120^\circ + \cos 40^\circ) \right) - \cos^3 20^\circ = k$$

$$\Rightarrow \cos 20^\circ \left(\cos^2 20^\circ - \frac{3}{2} \left(-\frac{1}{2} + \cos 40^\circ \right) \right) - \cos^3 20^\circ = k$$

$$\Rightarrow \cos^3 20^\circ - \frac{3}{2} \cos 20^\circ \left(-\frac{1}{2} + \cos 40^\circ \right) - \cos^3 20^\circ = k$$

$$\Rightarrow \frac{3}{4} \cos 20^\circ - \frac{3}{2} \cos 20^\circ \cos 40^\circ = k$$

$$\Rightarrow \frac{3}{4} \cos 20^\circ - \frac{3}{4} (\cos 60^\circ + \cos 20^\circ) = k$$

$$\Rightarrow \frac{3}{4} \cos 20^\circ - \frac{3}{4} \times \frac{1}{2} - \frac{3}{4} \cos 20^\circ = k$$

$$\Rightarrow k = -\frac{3}{8} \Rightarrow \frac{4k}{3} = \frac{-1}{2} = \cos \frac{2\pi}{3}$$

Question6

$$\cos 13^\circ \sin 17^\circ \sin 21^\circ \cos 47^\circ =$$

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Options:

A.

$$\frac{1}{32}(1 + \sqrt{2} - \sqrt{3})$$

B.

$$\frac{1}{16}(1 + \sqrt{3} + \sqrt{5})$$

C.

$$\frac{1}{16}(2 + \sqrt{3} - \sqrt{5})$$

D.

$$\frac{1}{32}(1 + 2\sqrt{3} - \sqrt{5})$$

Answer: D

Solution:

$$\begin{aligned}
& \cos 13^\circ \cdot \sin 17^\circ \cdot \sin 21^\circ \cdot \cos 47^\circ \\
&= (\cos 13^\circ \cos 47^\circ) \cdot (\sin 17^\circ \sin 21^\circ) \\
&= \frac{1}{2} [\cos (34^\circ) + \cos (60^\circ)] \cdot \frac{1}{2} [\cos (4^\circ) - \cos (38^\circ)] \\
&\quad [\because \sin A \sin B = \frac{1}{2} [\cos (A - B) - \cos (A + B)] \text{ and } \cos A \cos B = \frac{1}{2} [\cos (A - B) + \cos (A + B)]] \\
&= \frac{1}{4} \left[\cos 34^\circ + \frac{1}{2} \right] [\cos 4^\circ - \cos 38^\circ] \\
&= \frac{1}{4} [\cos 34^\circ \cos 4^\circ - \cos 34^\circ \cos 38^\circ + \frac{1}{2} \cos 4^\circ - \frac{1}{2} \cos 38^\circ] \\
&= \frac{1}{4} \left[\frac{1}{2} \{ \cos (30^\circ) + \cos (38^\circ) \} - \frac{1}{2} \{ \cos (4^\circ) + \cos (72^\circ) \} + \frac{1}{2} \cos 4^\circ - \frac{1}{2} \cos 38^\circ \right] \\
&= \frac{1}{4} \left[\frac{1}{2} \left(\frac{\sqrt{3}}{2} + \cos 38^\circ \right) - \frac{1}{2} (\cos 4^\circ + \cos 72^\circ) + \frac{1}{2} \cos 4^\circ - \frac{1}{2} \cos 38^\circ \right] \\
&= \frac{1}{8} \left[\frac{\sqrt{3}}{2} - \cos 72^\circ \right] = \frac{\sqrt{3}}{16} - \frac{\cos 72^\circ}{8} \\
&= \frac{\sqrt{3}}{16} - \frac{\sqrt{5}-1}{4} \times \frac{1}{8} \left(\because \cos 72^\circ = \frac{\sqrt{5}-1}{4} \right) \\
&= \frac{\sqrt{3}}{16} - \frac{\sqrt{5}-1}{32} = \frac{2\sqrt{3}-\sqrt{5}+1}{32} = \frac{1}{32} (2\sqrt{3}-\sqrt{5}+1)
\end{aligned}$$

Question 7

$$\sin \frac{\pi}{5} + \sin \frac{2\pi}{5} + \sin \frac{3\pi}{5} + \sin \frac{4\pi}{5} =$$

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Options:

A.

1

B.

$\sqrt{5}$

C.

$$\frac{1}{4}(\sqrt{5}+1)(\sqrt{10+2\sqrt{5}})$$

D.

$$\frac{1}{4}(\sqrt{5}-1)(\sqrt{10+2\sqrt{5}})$$

Answer: C

Solution:

$$\begin{aligned}
& \sin \frac{\pi}{5} + \sin \frac{2\pi}{5} + \sin \frac{3\pi}{5} + \sin \frac{4\pi}{5} \\
&= \sin \frac{\pi}{5} + \sin \frac{2\pi}{5} + \sin \left(\pi - \frac{2\pi}{5} \right) + \sin \left(\pi - \frac{\pi}{5} \right) \\
&= \sin \frac{\pi}{5} + \sin \frac{2\pi}{5} + \sin \frac{2\pi}{5} + \sin \frac{\pi}{5} = 2 \sin \frac{\pi}{5} + 2 \sin \frac{2\pi}{5} \\
&= 2 \left[\sin \frac{\pi}{5} + \sin \frac{2\pi}{5} \right] = 4 \sin \frac{3\pi}{10} \cos \frac{\pi}{10} \\
&= 4 \frac{(\sqrt{5}+1)}{4} \frac{(\sqrt{10+2\sqrt{5}})}{4} = \frac{1}{4}(\sqrt{5}+1)(\sqrt{10+2\sqrt{5}})
\end{aligned}$$

Question8

$$\operatorname{cosec} 48^\circ + \operatorname{cosec} 96^\circ + \operatorname{cosec} 192^\circ + \operatorname{cosec} 384^\circ =$$

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Options:

A.

$$4\sqrt{3}$$

B.

$$-4\sqrt{3}$$

C.

$$0$$

D.

$$1$$

Answer: C

Solution:

$$\operatorname{cosec} 48^\circ + \operatorname{cosec} 96^\circ + \operatorname{cosec} 192^\circ + \operatorname{cosec} 384^\circ$$

$$\because \operatorname{cosec} \theta = \cot \theta/2 - \cot \theta$$

$$\therefore \operatorname{cosec} 48^\circ = \cot 24^\circ - \cot 48^\circ$$

$$\operatorname{cosec} 96^\circ = \cot 48^\circ - \cot 96^\circ$$

$$\operatorname{cosec} 192^\circ = \cot 96^\circ - \cot 192^\circ$$

$$\operatorname{cosec} 384^\circ = \operatorname{cosec} 24^\circ$$

$$= \cot 24^\circ - \cot 48^\circ + \cot 48^\circ - \cot 96^\circ + \cot 96^\circ - \cot 192^\circ + \cot 24^\circ$$

$$\begin{aligned}
&= \cot 24^\circ - \cot 192^\circ + \cos 24^\circ \\
&\text{and } \operatorname{cosec} 24^\circ = \cot 12^\circ - \cot 24^\circ \\
&= \cot 24^\circ - \cot 192^\circ + \cot 12^\circ - \cot 24^\circ \\
&= \cot 12^\circ - \cot 192^\circ \\
&= \cot 12^\circ - \cot (180 + 12^\circ) \\
&= \cot 12^\circ - \cot 12^\circ = 0
\end{aligned}$$

Question9

If $\cos \theta = -\frac{3}{5}$ and θ does not lie in second quadrant, then $\tan \frac{\theta}{2} =$

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Options:

A.

2

B.

1

C.

-2

D.

-1

Answer: C

Solution:

$\because \cos \theta = -\frac{3}{5} < 0$ and θ does not lie in second quadrant.

Thus, θ lies in third quadrant.

So, $\theta/2$ must lie in second quadrant.

There $\tan \theta/2$ must be negative.

then, $\cos \theta = \frac{3}{5} = \frac{B}{H}$

$\therefore \tan \theta = \frac{4}{3}$

So, $\tan \theta = \frac{2 \tan \theta/2}{1 - \tan^2 \theta/2}$

let $\tan \theta/2 = x$

$$\begin{aligned}\Rightarrow \quad \frac{4}{3} &= \frac{2x}{1-x^2} \Rightarrow 4 - 4x^2 = 6x \\ \Rightarrow \quad 2x^2 + 3x - 2 &= 0 \\ \Rightarrow \quad 2x^2 + 4x - x - 2 &= 0 \\ \Rightarrow \quad (2x - 1)(x + 2) &= 0\end{aligned}$$

Therefore,

$$\begin{aligned}x &= 1/2 \text{ or } x = -2 \\ \Rightarrow \quad \tan \theta/2 &= 1/2 \text{ or } \tan \theta/2 = -2\end{aligned}$$

$\therefore \tan \theta/2$ is negative

Hence, $\tan \theta/2 = -2$

Question10

If α is the maximum value and β is the minimum value of $\cos^2 \frac{x}{4} + \sin \frac{x}{4}$, $x \in R$, then $\alpha - \beta =$

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Options:

A.

$$\frac{1}{4}$$

B.

$$\frac{9}{4}$$

C.

$$2$$

D.

$$3$$

Answer: B

Solution:

$$\begin{aligned}
 \text{Let } f(x) &= \cos^2\left(\frac{x}{4}\right) + \sin\left(\frac{x}{4}\right) \\
 &= 1 - \sin^2\left(\frac{x}{4}\right) + \sin\left(\frac{x}{4}\right) \\
 &= -\sin^2\left(\frac{x}{4}\right) + \sin\left(\frac{x}{4}\right) + 1 \\
 &= \frac{5}{4} - \left(\sin\left(\frac{x}{4}\right) - \frac{1}{2}\right)^2 \\
 \therefore f(x)|_{\max} &= \alpha = \frac{5}{4} \text{ when } \sin\frac{x}{4} = \frac{1}{2} \\
 \text{and } f(x)|_{\min} &= \beta = -1 \text{ when } \sin\frac{x}{4} = -1 \\
 \therefore \alpha - \beta &= \frac{5}{4} - (-1) = \frac{9}{4}
 \end{aligned}$$

Question11

If A and B are positive acute angles satisfying $3 \cos^2 A + 2 \cos^2 B = 4$ and $\frac{3 \sin A}{\sin B} = \frac{2 \cos B}{\cos A}$, then $A + 2B =$

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Options:

A.

30°

B.

45°

C.

60°

D.

90°

Answer: D

Solution:

$$\begin{aligned}
 3 \cos^2 A + 2 \cos^2 B &= 4 \\
 \Rightarrow 3(1 - \sin^2 A) + 2(1 - \sin^2 B) &= 4
 \end{aligned}$$

Now, $3 \sin A \cdot \cos A = 2 \sin B \cdot \cos B$ (given)

$$\Rightarrow 3 \sin 2A = 2 \sin 2B \quad \dots (i)$$

$$\begin{aligned}
 \cos(A + 2B) &= \cos A \cos 2B - \sin A \sin 2B \\
 &= \cos A \cdot 3 \sin^2 A - \sin A \cdot \frac{3}{2} \cdot 2 \sin A \cos A \quad (\text{from Eq. (i)}) \\
 &= 3 \cos A \sin^2 A - 3 \cos A \sin^2 A = 0 \\
 \therefore A + 2B &= 90^\circ
 \end{aligned}$$

Question 12

If $\sin x - \sin y = \frac{27}{65}$ and $\cos x - \cos y = \frac{-21}{65}$, then $\sin(x + y) =$

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Options:

A.

$$-\frac{63}{65}$$

B.

$$\frac{16}{65}$$

C.

$$\frac{63}{65}$$

D.

$$-\frac{16}{65}$$

Answer: C

Solution:

$$\begin{aligned}
 &\text{Given, } \sin x - \sin y = 27/65 \text{ and} \\
 \cos x - \cos y &= \frac{-21}{65} \\
 2 \sin \left(\frac{x-y}{2} \right) \cos \left(\frac{x+y}{2} \right) &= \frac{27}{65} \text{ and} \\
 2 \sin \left(\frac{x+y}{2} \right) \sin \left(\frac{x-y}{2} \right) &= \frac{21}{65} \\
 \therefore \tan \left(\frac{x+y}{2} \right) &= \frac{21}{27} = \frac{7}{9} \\
 \sin(x+y) &= \frac{2 \tan \left(\frac{x+y}{2} \right)}{1 + \tan^2 \frac{x+y}{2}} = \frac{2 \times \frac{7}{9}}{1 + \frac{49}{81}} \\
 \sin(x+y) &= \frac{63}{65}
 \end{aligned}$$

Question13

If α, β are the acute angles such that $\frac{\sin \alpha}{\sin \beta} = \frac{6}{5}$ and $\frac{\cos \alpha}{\cos \beta} = \frac{9}{5\sqrt{5}}$, then $\sin \alpha =$

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Options:

A.

$$\frac{4}{5}$$

B.

$$\frac{3}{5}$$

C.

$$\frac{3}{4}$$

D.

$$\frac{2}{3}$$

Answer: A

Solution:

$$\begin{aligned}\sin \beta &= \frac{5}{6} \sin \alpha \\ \Rightarrow \cos \beta &= \frac{5\sqrt{5}}{9} \cos \alpha \\ \Rightarrow \sin^2 \beta + \cos^2 \beta &= 1 \\ \Rightarrow \left(\frac{5}{6} \sin \alpha \right)^2 + \left(\frac{5\sqrt{5}}{9} \cos \alpha \right)^2 &= 1 \\ \Rightarrow \frac{25}{36} \sin^2 \alpha + \frac{125}{81} \cos^2 \alpha &= 1 \\ \Rightarrow \frac{25}{36} \sin^2 \alpha + \frac{125}{81} - \frac{125}{81} \sin^2 \alpha &= 1 \\ \Rightarrow \left(\frac{225 - 500}{324} \right) \sin^2 \alpha &= \frac{81 - 125}{81} \\ \Rightarrow \sin^2 \alpha &= \frac{-44}{81} \times \frac{324}{-275} \\ \Rightarrow \sin^2 \alpha &= \frac{4}{9} \times \frac{11}{9} \times \frac{4}{25} \times \frac{81}{11} \\ \Rightarrow \sin^2 \alpha &= \frac{16}{25} \Rightarrow \sin \alpha = \frac{4}{5}\end{aligned}$$

Question14

If $\left(\frac{\sin 3\theta}{\sin \theta}\right)^2 - \left(\frac{\cos 3\theta}{\cos \theta}\right)^2 = a \cos b\theta$, then $a : b =$

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Options:

A.

4 : 1

B.

8 : 1

C.

3 : 2

D.

2 : 1

Answer: A

Solution:

$$\begin{aligned} \left(\frac{\sin 3\theta}{\sin \theta}\right)^2 - \left(\frac{\cos 3\theta}{\cos \theta}\right)^2 &= a \cos b\theta \\ \Rightarrow \left(\frac{3 \sin \theta - 4 \sin^3 \theta}{\sin \theta}\right) - \left(\frac{4 \cos^3 \theta - 3 \cos \theta}{\cos \theta}\right) &= a \cos b\theta \\ \Rightarrow (3 - 4 \sin^2 \theta)^2 - (4 \cos^2 \theta - 3)^2 &= a \cos b\theta \\ \Rightarrow 9 + 16 \sin^4 \theta - 24 \sin^2 \theta - 16 \cos^4 \theta - 9 + 24 \cos^2 \theta &= a \cos b\theta \\ \Rightarrow 24 (\cos^2 \theta - \sin^2 \theta) - 16 (\cos^4 \theta - \sin^4 \theta) &= a \cos b\theta \\ \Rightarrow 8 (\cos^2 \theta - \sin^2 \theta) (3 - 2 (\cos^2 \theta + \sin^2 \theta)) &= a \cos b\theta, \\ \Rightarrow 8 \cos 2\theta (3 - 2) &= a \cos b\theta \\ \Rightarrow 8 \cos 2\theta &= a \cos b\theta \Rightarrow a = 8, b = 2 \\ \therefore a : b &= 8 : 2 = 4 : 1 \end{aligned}$$

Question15

An aeroplane is flying at a constant speed, parallel to the horizontal ground at a height of 5 kms . A person on the ground observed that the angle of elevation of the plane is changed from 15° to 30° in the duration of 50 seconds, then the speed of the plane (in kmph) is

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Options:

A.

100

B.

720

C.

360

D.

540

Answer: B

Solution:

Let x_1 be the initial horizontal distance

$$\tan 15^\circ = \frac{5}{x_1} \Rightarrow x_1 = \frac{5}{\tan 15^\circ}$$

Let x_2 be the final horizontal distance

$$\begin{aligned}\tan 30^\circ &= \frac{5}{x_2} \\ \Rightarrow x_2 &= \frac{5}{\tan 30^\circ}\end{aligned}$$

The distance travelled is $d = x_1 - x_2$

$$\begin{aligned}d &= \frac{5}{\tan 15^\circ} - \frac{5}{\tan 30^\circ} \\ d &\approx 5(3.732 - 1.732) \\ d &\approx 5 \times 2 \\ d &= 10 \text{ km}\end{aligned}$$

$$\begin{aligned}\text{speed} &= \frac{d}{t} = \frac{10}{50} = 0.2 \text{ km/s} \\ &= 3600 \times 0.2 \text{ km/h} = 720 \text{ km/h} \\ \Rightarrow \text{The speed of the plane is } 720 \text{ km/h}\end{aligned}$$

Question 16

If $A + B = \frac{\pi}{4}$, then $\frac{\cos B - \sin B}{\cos B + \sin B} =$

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Options:

A.

$$\sin A$$

B.

$$\cos A$$

C.

$$\tan A$$

D.

$$\cot A$$

Answer: C

Solution:

We have,

$$\begin{aligned} A + B &= \frac{\pi}{4} \\ \Rightarrow \frac{\cos B - \sin B}{\cos B + \sin B} &= \frac{\sin\left(\frac{\pi}{4} + A\right) - \sin\left(\frac{\pi}{4} - A\right)}{\sin\left(\frac{\pi}{4} + A\right) + \sin\left(\frac{\pi}{4} - A\right)} \\ &= \frac{2 \cos \frac{\pi}{4} \sin A}{2 \sin \frac{\pi}{4} \cos A} = \tan A \end{aligned}$$

Question17

If $7 \cos \theta - \sin \theta = 5$ and $\tan \theta > 0$, then $\tan \theta =$

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Options:

A.

$$\frac{7}{12}$$

B.

$$\frac{3}{4}$$

C.

$$\frac{4}{3}$$

D.

$$\frac{12}{7}$$

Answer: B

Solution:

$$7 \cos \theta - \sin \theta = 5, \tan \theta > 0$$

$$7 - \tan \theta = 5 \sec \theta$$

Squaring both sides, we get

$$\Rightarrow 49 + \tan^2 \theta - 14 \tan \theta = 25 (1 + \tan^2 \theta)$$

$$\Rightarrow 24 \tan^2 \theta + 14 \tan \theta - 24 = 0$$

$$\Rightarrow 12 \tan^2 \theta + 7 \tan \theta - 12 = 0$$

$$\Rightarrow 12 \tan^2 \theta + 16 \tan \theta - 9 \cdot \tan \theta - 12 = 0$$

$$\Rightarrow (3 \tan \theta + 4)(4 \tan \theta - 3) = 0$$

$$\tan \theta = \frac{3}{4}$$

Question18

$$\sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ =$$

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Options:

A.

$$\frac{-3}{8}$$

B.

$$\frac{3}{4}$$

C.

$$\frac{\sqrt{3}}{2}$$

D.

$$\frac{-1}{3}$$

Answer: A

Solution:

$$\begin{aligned}
& \sin^3 10^\circ + \sin^3 50^\circ - \sin^3 70^\circ \\
\Rightarrow & \sin 3x = 3 \sin x - 4 \sin^3 x \\
\Rightarrow & \sin^3 x = \frac{1}{4}(3 \sin x - \sin 3x) \\
\Rightarrow & \frac{1}{4}[3 \sin 10^\circ - \sin 30^\circ + 3 \sin 50^\circ - \sin 150^\circ - 3 \sin 70^\circ + \sin 210^\circ] \\
\Rightarrow & \frac{1}{4}\left[3 \sin 10^\circ + 3 \sin 50^\circ - 3 \sin 70^\circ - \frac{1}{2} - \frac{1}{2} - \frac{1}{2}\right] \\
\Rightarrow & \frac{3}{4}\left[\sin 10^\circ + \sin 50^\circ - \sin 70^\circ - \frac{1}{2}\right] \\
\Rightarrow & \frac{3}{4}\left[2 \sin 30^\circ \cos 20^\circ - \sin 70^\circ - \frac{1}{2}\right] \\
\Rightarrow & \frac{3}{4}\left[\cos 20^\circ - \sin 70^\circ - \frac{1}{2}\right] \\
\Rightarrow & \frac{3}{4}\left[\sin 70^\circ - \sin 70^\circ - \frac{1}{2}\right] = -\frac{3}{8}
\end{aligned}$$

Question19

$$\frac{1}{\sin 1^\circ \sin 2^\circ} + \frac{1}{\sin 2^\circ \sin 3^\circ} + \frac{1}{\sin 3^\circ \sin 4^\circ} + \frac{1}{\sin 89^\circ \sin 90^\circ} =$$

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Options:

A.

$$\frac{\sin 1^\circ}{\tan 1^\circ}$$

B.

$$\frac{1}{\sin^2 \varphi}$$

C.

$$\frac{\cot 1^\circ}{\sin 1^\circ}$$

D.

$$\frac{\tan 1^\circ}{\cos 1^\circ}$$

Answer: C

Solution:

$$\begin{aligned}
&\text{We have, } \sum_{r=1}^{89} \frac{1}{\sin r^\circ \cdot \sin(r+1)^\circ} \\
&\Rightarrow \frac{1}{\sin 1^\circ} \sum_{r=1}^{89} \frac{\sin((r^\circ + 1^\circ) - r^\circ)}{\sin r^\circ \cdot \sin(r^\circ + 1^\circ)} \\
&\Rightarrow \frac{1}{\sin 1^\circ} \sum_{r=1}^{89} \frac{\sin(r+1)^\circ \cos r^\circ}{\sin r^\circ \cdot \sin(r+1)^\circ} \\
&\Rightarrow \frac{1}{\sin 1^\circ} \sum_{r=1}^{89} \cot r^\circ - \cot(r+1)^\circ \\
&\Rightarrow \frac{1}{\sin 1^\circ} [\cot 1^\circ - \cot 90^\circ] = \frac{\cot 1^\circ}{\sin 1^\circ}
\end{aligned}$$

Question20

$$\cos^3 \frac{\pi}{8} \cos \frac{3\pi}{8} + \sin^3 \frac{\pi}{8} \sin \frac{3\pi}{8} =$$

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Options:

A.

$$\frac{1}{2\sqrt{2}}$$

B.

$$\frac{1}{2}$$

C.

$$\frac{1}{\sqrt{2}}$$

D.

$$\frac{1}{4}$$

Answer: A

Solution:

Given,

$$\cos^3 \frac{\pi}{8} \cdot \cos \frac{3\pi}{8} + \sin^3 \frac{\pi}{8} \cdot \sin \frac{3\pi}{8}$$

$$\text{Using } \frac{3\pi}{8} = \frac{\pi}{2} - \frac{\pi}{8}$$

$$\Rightarrow \cos \frac{3\pi}{8} = \sin \frac{\pi}{8} \text{ and } \sin \frac{3\pi}{8} = \cos \frac{\pi}{8}$$

$$\begin{aligned} \Rightarrow \cos^3 \frac{\pi}{8} \cdot \sin \frac{\pi}{8} + \sin^3 \frac{\pi}{8} \cdot \cos \frac{\pi}{8} \\ = \sin \frac{\pi}{8} \cdot \cos \frac{\pi}{8} \left(\cos^2 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} \right) \\ = \frac{1}{2} 2 \sin \frac{\pi}{8} \cos \frac{\pi}{8} \\ = \frac{1}{2} \sin \frac{\pi}{4} = \frac{1}{2\sqrt{2}} \end{aligned}$$

Question21

If $A + B + C = \frac{\pi}{4}$, then $\sin 4A + \sin 4B + \sin 4C =$

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Options:

A.

$$4 \cos 2A \cos 2B \cos 2C$$

B.

$$4 \sin 2A \sin 2B \sin 2C$$

C.

$$1 + 4 \sin 2A \sin 2B \sin 2C$$

D.

$$1 + 4 \cos 2A \cos 2B \cos 2C$$

Answer: A

Solution:

Given, $A + B + C = \frac{\pi}{4}$

$\therefore 2A + 2B + 2C = \frac{\pi}{2}$

Now, $\sin 4A + \sin 4B + \sin 4C$

$2 \sin(2A + 2B) \cdot \cos(2A - 2B) + 2 \sin 2C$

$= 2 \sin\left(\frac{\pi}{2} - 2C\right) \cdot \cos 2C + 2 \sin 2C \cdot \cos 2C$

$= 2 \cos 2C \cdot \cos(2A - 2B) + 2 \sin 2C \cdot \cos 2C$

$= 2 \cos 2C \{\cos(2A - 2B) + \sin 2C\}$

$= 2 \cos 2C [\cos(2A - 2B) + \cos(2A + 2B)]$

$= 2 \cos 2C [2 \cos 2A \cdot \cos 2B]$

$= 4 \cos 2A \cdot \cos 2B \cdot \cos 2C$

Question 22

If $630^\circ < \theta < 810^\circ$ and $\tan \theta = -\frac{7}{24}$, then $\cos\left(\frac{\theta}{4}\right) =$

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Options:

A.

$$-\sqrt{\frac{7+5\sqrt{2}}{10\sqrt{2}}}$$

B.

$$\sqrt{\frac{7+5\sqrt{2}}{2\sqrt{2}}}$$

C.

$$-\sqrt{\frac{5\sqrt{2}-7}{10\sqrt{2}}}$$

D.

$$\sqrt{\frac{5\sqrt{2}-7}{2\sqrt{2}}}$$

Answer: A

Solution:

Given, $630^\circ < \theta < 810^\circ$ and

$$\tan \theta = -\frac{7}{24}$$

$\therefore \theta$ lies IV quadrant

$$\begin{aligned}
\therefore \cos \theta &= \frac{24}{25} \quad 315^\circ < \frac{\theta}{2} < 405^\circ \\
2 \cos^2 \frac{\theta}{2} - 1 &= \frac{24}{25} \quad 157\frac{1}{2} < \frac{\theta}{4} < 2.02\frac{1}{2} \\
2 \cos^2 \frac{\theta}{2} &= \frac{49}{25} \quad \frac{\theta}{4} \text{ lies III quadrant} \\
\Rightarrow \cos \frac{\theta}{2} &= \frac{7}{5\sqrt{2}} \Rightarrow 2 \cos^2 \frac{\theta}{4} - 1 = \frac{7}{5\sqrt{2}} \\
\Rightarrow 2 \cos^2 \frac{\theta}{4} &= 1 + \frac{7}{5\sqrt{2}} = \frac{7 + 5\sqrt{2}}{5\sqrt{2}} \\
\Rightarrow \cos \frac{\theta}{4} &= -\sqrt{\frac{7 + 5\sqrt{2}}{10\sqrt{2}}}
\end{aligned}$$

Question23

For $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ if $2 \cos \theta + \sin \theta = 1$ and $7 \cos \theta + 6 \sin \theta = k$, then the possible values of k are

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Options:

A.

8,-2

B.

6,2

C.

12,4

D.

7,6

Answer: B

Solution:

We have,

$$2 \cos \theta + \sin \theta = 1 \quad \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Squaring both sides, we get

$$\begin{aligned}
&\Rightarrow (2 \cos \theta)^2 = (1 - \sin \theta)^2 \\
&\Rightarrow 4 \cos^2 \theta = 1 + \sin^2 \theta - 2 \sin \theta \\
&\Rightarrow 5 \sin^2 \theta - 2 \sin \theta - 3 = 0 \\
&\Rightarrow (\sin \theta - 1)(5 \sin \theta + 3) = 0 \\
&\Rightarrow \sin \theta = 1 \text{ and } -\frac{3}{5} \\
&\Rightarrow \cos \theta = 0 \text{ and } +\frac{4}{5} \\
&\therefore 7 \cos \theta + 6 \sin \theta = k, \text{ when } \sin \theta = 1 \\
&\Rightarrow 7(0) + 6(1) = k \Rightarrow k = 6
\end{aligned}$$

$$\begin{aligned}
&\text{When, } \sin \theta = -\frac{3}{5} \text{ and } \cos \theta = \frac{4}{5} \\
&k = 7 \left(\frac{4}{5} \right) - 6 \left(\frac{3}{5} \right) = \frac{28 - 18}{5} = \frac{10}{5} = 2
\end{aligned}$$

Question24

$$\sum_{k=0}^{12} \frac{1}{\sin\left((k+1)\frac{\pi}{6} + \frac{\pi}{4}\right) \sin\left(\frac{k\pi}{6} + \frac{\pi}{4}\right)} =$$

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Options:

A.

$$2(\sqrt{3} + 1)$$

B.

$$2(3 - \sqrt{3})$$

C.

$$2(2 - \sqrt{3})$$

D.

$$2(\sqrt{3} - 1)$$

Answer: D

Solution:

$$\sum_{k=0}^{12} \frac{1}{\sin \left((k+1) \frac{\pi}{6} + \frac{\pi}{4} \right) \cdot \sin \left(\frac{k\pi}{6} + \frac{\pi}{4} \right)}$$

Multiplying N^r and D^r by $\sin \pi/6$,

$$\frac{1}{\sin \frac{\pi}{6}} \sum_{k=0}^{12} \frac{\sin \left\{ \left((k+1) \frac{\pi}{6} + \frac{\pi}{4} \right) - \left(\frac{k\pi}{6} + \frac{\pi}{4} \right) \right\}}{2 \sum_{k=0}^{12} \left((k+1) \frac{\pi}{6} + \frac{\pi}{4} \right) \cdot \sin \left(\frac{k\pi}{6} + \frac{\pi}{4} \right)}$$

$$= 2 \left[\cot \left(\frac{k\pi}{6} + \frac{\pi}{4} \right) - \cot \left((k+1) \frac{\pi}{6} + \frac{\pi}{4} \right) \right]$$

$$+ \left\{ \cot \left(\frac{\pi}{6} + \frac{\pi}{4} \right) - \cot \left(\frac{\pi}{6} + \frac{\pi}{4} \right) \right\}$$

$$+ \dots + \left\{ \cot \left(\frac{12\pi}{6} + \frac{\pi}{4} \right) - \cot \left(\frac{13\pi}{6} + \frac{\pi}{4} \right) \right\}$$

$$= 2 \left[\cot \frac{\pi}{4} + \cot \left(\frac{\pi}{6} + \frac{\pi}{4} \right) \right]$$

$$= 2 \left[1 - \frac{\sqrt{3}-1}{\sqrt{3}+1} \right] = 2 \left(\frac{2}{\sqrt{3}+1} \right) = \frac{4(\sqrt{3}-1)}{2}$$

$$= 2(\sqrt{3}-1)$$

Question25

If $\cos \alpha = \sec h\beta$, then $\beta =$

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Options:

A.

$$\log(\sec \alpha + \tan \alpha)$$

B.

$$\log(\sec \alpha - \tan \alpha)$$

C.

$$\log(\sin \alpha + \cos \alpha)$$

D.

$$\log(\cos \alpha + \cot \alpha)$$

Answer: A

Solution:

$$\text{Given } \cos \alpha = \sec h\beta$$

$$\text{or } \cos h\beta = \sec \alpha$$

$$\Rightarrow \beta = \cos h^{-1}(\sec \alpha)$$

$$\Rightarrow \beta = \log \left(\sec \alpha + \sqrt{\sec^2 \alpha - 1} \right)$$

$$\left[\because \cos h^{-1} x = \log \left(x + \sqrt{x^2 - 1} \right) \right]$$

$$\beta = \log(\sec \alpha + \tan \alpha)$$

Question 26

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{2\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{4\pi}{16} \\ + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{6\pi}{16} + \tan^2 \frac{7\pi}{16} \text{ is equal to}$$

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Options:

A. 35

B. 41

C. 37

D. 33

Answer: A

Solution:

$$\begin{aligned} & \tan^2 \frac{\pi}{16} + \tan^2 \frac{2\pi}{16} + \tan^2 \frac{3\pi}{16} \\ & \quad + \tan^2 \frac{4\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{6\pi}{16} \\ + \tan^2 \frac{7\pi}{16} &= \left(\sec^2 \frac{\pi}{16} - 1 \right) + \left(\sec^2 \frac{2\pi}{16} - 1 \right) \\ & \quad + \left(\sec^2 \frac{3\pi}{16} - 1 \right) + 1 + \tan^2 \left(\frac{\pi}{2} - \frac{3\pi}{16} \right) \\ & \quad + \tan^2 \left(\frac{\pi}{2} - \frac{2\pi}{16} \right) + \tan^2 \left(\frac{\pi}{2} - \frac{\pi}{16} \right) \\ &= \left(\sec^2 \frac{\pi}{16} - 1 \right) + \left(\sec^2 \frac{2\pi}{16} - 1 \right) + \left(\sec^2 \frac{3\pi}{16} - 1 \right) + 1 \\ & \quad + \cot^2 \left(\frac{3\pi}{16} \right) + \cot^2 \left(\frac{2\pi}{16} \right) + \cot^2 \left(\frac{\pi}{16} \right) \\ &= \left(\sec^2 \frac{\pi}{16} - 1 \right) + \left(\sec^2 \frac{2\pi}{16} - 1 \right) \\ & \quad + \left(\sec^2 \frac{3\pi}{16} - 1 \right) + 1 + \left(\operatorname{cosec}^2 \frac{3\pi}{16} - 1 \right) \\ & \quad + \left(\operatorname{cosec}^2 \frac{2\pi}{16} - 1 \right) + \left(\operatorname{cosec}^2 \frac{\pi}{16} - 1 \right) \\ &= \left(\sec^2 \frac{\pi}{16} + \operatorname{cosec}^2 \frac{\pi}{16} \right) + \left(\sec^2 \frac{2\pi}{16} + \operatorname{cosec}^2 \frac{2\pi}{16} \right) + \left(\sec^2 \frac{3\pi}{16} + \operatorname{cosec}^2 \frac{3\pi}{16} \right) - 6 + 1 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sin^2 \frac{\pi}{16} \cos^2 \frac{\pi}{16}} + \frac{1}{\sin^2 \frac{2\pi}{16} \cos^2 \frac{2\pi}{16}} \\
&+ \frac{1}{\sin^2 \frac{3\pi}{16} \cos^2 \frac{3\pi}{16}} - 5 \\
&= \frac{4}{\sin^2 \frac{\pi}{8}} + \frac{4}{\sin^2 \frac{\pi}{4}} + \frac{4}{\sin^2 \frac{3\pi}{8}} - 5 \\
&= 4 \left(\frac{1}{\sin^2 \frac{\pi}{8}} + \frac{1}{\cos^2 \frac{\pi}{8}} \right) + 4 \times 2 - 5 \\
&= 4 \left(\frac{1}{\sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8}} \right) + 3 = \frac{4 \times 4}{\sin^2 \frac{\pi}{4}} + 3 \\
&= 2 \times 16 + 3 = 35
\end{aligned}$$

Question 27

$\sin^2 18^\circ + \sin^2 24^\circ + \sin^2 36^\circ + \sin^2 42^\circ + \sin^2 78^\circ$
 $+ \sin^2 90^\circ + \sin^2 96^\circ + \sin^2 102^\circ + \sin^2 138^\circ + \sin^2 162^\circ$ is
 equal to

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Options:

- A. $\frac{11}{2}$
- B. $\frac{9}{2}$
- C. 5
- D. 4

Answer: A

Solution:

$$\begin{aligned}
&\sin^2 18^\circ + \sin^2 24^\circ + \sin^2 36^\circ + \sin^2 42^\circ + \sin^2 78^\circ + \sin^2 90^\circ \\
&+ \sin^2 96^\circ + \sin^2 102^\circ + \sin^2 138^\circ + \sin^2 162^\circ \\
&\because \sin^2 \theta = \sin^2 (180^\circ - \theta)
\end{aligned}$$

$$\begin{aligned}
& \therefore \sin^2 18^\circ + \sin^2 24^\circ + \sin^2 36^\circ + \sin^2 42^\circ + \sin^2 78^\circ + 1 + \sin^2 84^\circ + \sin^2 78^\circ + \sin^2 42^\circ + \sin^2 18^\circ \\
&= 2 \sin^2 18^\circ + 2 \sin^2 42^\circ + 2 \sin^2 78^\circ + \sin^2 24^\circ + \sin^2 36^\circ + \sin^2 84^\circ + 1 \\
&= 2 \sin^2 18^\circ + 2 \sin^2 42^\circ + 2 \cos^2 12^\circ + \sin^2 24^\circ + \sin^2 36^\circ + \cos^2 6^\circ + 1 \\
&= 2 [\cos^2 12^\circ + \sin^2 (30^\circ - 12^\circ) + \sin^2 (30^\circ + 12^\circ)] \\
&\quad + [\cos^2 6^\circ + \sin^2 (30^\circ + 6^\circ) + \sin^2 (30^\circ - 6^\circ)] + 1 \\
&\therefore \cos^2 \theta + \sin^2 (30^\circ + \theta) + \sin^2 (30^\circ - \theta) = 0
\end{aligned}$$

$$\begin{aligned}
& \cos^2 \theta + [\sin 30^\circ \cos \theta + \cos 30^\circ \sin \theta]^2 \\
& \quad + [\sin 30^\circ \cos \theta - \cos 30^\circ \sin \theta]^2
\end{aligned}$$

$$\text{Using } (a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$$

$$\Rightarrow \cos^2 \theta + 2 [\sin^2 30^\circ \cos^2 \theta + \cos^2 30^\circ \sin^2 \theta]$$

$$\Rightarrow \cos^2 \theta + 2 \left[\frac{1}{4} \cos^2 \theta + \frac{3}{4} \sin^2 \theta \right]$$

$$\Rightarrow \cos^2 \theta + \frac{1}{2} \cos^2 \theta + \frac{3}{2} \sin^2 \theta$$

$$\Rightarrow 3/2 \quad [\because \cos^2 \theta + \sin^2 \theta = 1]$$

Using this result in Eq. (i), we get

$$2 \left(\frac{3}{2} \right) + \frac{3}{2} + 1 = \frac{11}{2}$$

Question28

If A and B are the angles of a triangle, then $\frac{\sin A + \sin B + \sin C}{\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} - 1}$ is equal to

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Options:

A. $-2 \tan \frac{B}{2}$

B. $-2 \cot \frac{B}{2}$

C. $2 \tan \frac{B}{2}$

D. $2 \cot \frac{B}{2}$

Answer: B

Solution:

We have,

$$\begin{aligned}
 & \frac{\sin A + \sin B + \sin C}{\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} - 1} \\
 &= \frac{2 \sin \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right) + 2 \sin \frac{B}{2} \cos \frac{B}{2}}{\sin^2 \frac{A}{2} - \cos^2 \frac{C}{2} + \sin^2 \frac{B}{2}} \\
 &= \frac{2 \cos \frac{B}{2} \cos \frac{A-C}{2} + 2 \sin \frac{B}{2} \cos \frac{B}{2}}{-\cos \left(\frac{A+C}{2} \right) \cos \left(\frac{A-C}{2} \right) + \sin^2 \frac{B}{2}} \\
 & [\because \sin^2 A - \cos^2 B = -\cos(A+B) \cos(A-B)] \\
 &= \frac{2 \cos \frac{B}{2} \left[\cos \left(\frac{A-C}{2} \right) + \sin \frac{B}{2} \right]}{-\sin \frac{B}{2} \left[\cos \left(\frac{A-C}{2} \right) + \sin \frac{B}{2} \right]} \\
 &= -2 \cot \frac{B}{2}
 \end{aligned}$$

Question29

If $\cos \alpha + 4 \cos \beta + 9 \cos \gamma = 0$ and $\sin \alpha + 4 \sin \beta + 9 \sin \gamma = 0$, then $81 \cos(2\gamma - 2\alpha) - 16 \cos(2\beta - 2\alpha)$ is equal to

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Options:

- A. $1 + 8 \cos(\beta - \alpha)$
- B. $\cos(\beta - \alpha)$
- C. $1 - 36 \cos(\beta - \alpha)$
- D. $1 + 6 \cos(\beta - \alpha)$

Answer: A

Solution:

We start with the given equations:

$$\begin{aligned}
 \cos \alpha + 4 \cos \beta + 9 \cos \gamma &= 0 \quad \dots (i) \\
 \sin \alpha + 4 \sin \beta + 9 \sin \gamma &= 0 \quad \dots (ii)
 \end{aligned}$$

Combine these equations using the complex form of the equations. Multiply both expressions:

$$\cos \theta + i \sin \theta = e^{i\theta}$$

Thus, combining both equations, we have:

$$\begin{aligned}
 (\cos \alpha + i \sin \alpha) + 4(\cos \beta + i \sin \beta) + 9(\cos \gamma + i \sin \gamma) &= 0 \\
 e^{i\alpha} + 4e^{i\beta} + 9e^{i\gamma} &= 0 \\
 \Rightarrow 1 + 4e^{i(\beta-\alpha)} &= -9e^{i(\gamma-\alpha)}
 \end{aligned}$$

Substitute using Euler's formula:

$$1 + [4 \cos(\beta - \alpha) + 4i \sin(\beta - \alpha)] = -[9 \cos(\gamma - \alpha) + 9i \sin(\gamma - \alpha)]$$

Next, square both sides and compare the real parts:

$$\begin{aligned} & [1 + 4 \cos(\beta - \alpha)]^2 - 16 \sin^2(\beta - \alpha) \\ &= [9 \cos(\gamma - \alpha)]^2 - [9 \sin(\gamma - \alpha)]^2 \\ &= 1 + 8 \cos(\beta - \alpha) + 16[\cos^2(\beta - \alpha) - \sin^2(\beta - \alpha)] \\ &= 81[\cos^2(\gamma - \alpha) - \sin^2(\gamma - \alpha)] \end{aligned}$$

Use the identities for double angles:

$$\begin{aligned} &\Rightarrow 81 \cos 2(\gamma - \alpha) - 16 \cos 2(\beta - \alpha) \\ &= 1 + 8 \cos(\beta - \alpha) \end{aligned}$$

Therefore, the expression $81 \cos(2\gamma - 2\alpha) - 16 \cos(2\beta - 2\alpha)$ simplifies to:

$$1 + 8 \cos(\beta - \alpha)$$

Question30

$\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha$ is equal to

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Options:

A. $\sin \alpha$

B. $\cos \alpha$

C. $\tan \alpha$

D. $\cot \alpha$

Answer: D

Solution:

We have,

$$\begin{aligned} & \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha \\ &= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \times \frac{1 - \tan^2 4\alpha}{2 \tan 4\alpha} \\ &= \tan \alpha + 2 \tan 2\alpha + \frac{8 \tan^2 4\alpha + 8 - 8 \tan^2 4\alpha}{2 \tan 4\alpha} \\ &= \tan \alpha + 2 \tan 2\alpha + \frac{8}{2 \left(\frac{2 \tan 2\alpha}{1 - \tan^2 2\alpha} \right)} \\ &= \tan \alpha + 2 \tan 2\alpha + \frac{4(1 - \tan^2 2\alpha)}{2 \tan 2\alpha} \\ &= \tan \alpha + \frac{4 \tan^2 2\alpha + 4 - 4 \tan^2 2\alpha}{2 \tan 2\alpha} \\ &= \tan \alpha + \frac{2}{\tan 2\alpha} = \tan \alpha + \frac{2(1 - \tan^2 \alpha)}{2 \tan \alpha} \\ &= \frac{2 \tan^2 \alpha + 2 - 2 \tan^2 \alpha}{2 \tan \alpha} = \frac{1}{\tan \alpha} = \cot \alpha \end{aligned}$$

Question31

$\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to

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Options:

A. 4

B. 3

C. 2

D. 1

Answer: A

Solution:

$$\begin{aligned} & \tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ \\ &= (\tan 9^\circ + \tan 81^\circ) - (\tan 27^\circ + \tan 63^\circ) \\ &= (\tan 9^\circ + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ) \\ &= \left(\frac{\cos 9^\circ}{\sin 9^\circ} + \frac{\sin 9^\circ}{\cos 9^\circ} \right) - \left(\frac{\sin 27^\circ}{\cos 27^\circ} + \frac{\cos 27^\circ}{\sin 27^\circ} \right) \\ &= \frac{\cos^2 9^\circ + \sin^2 9^\circ}{\sin 9^\circ \cdot \cos 9^\circ} - \frac{\sin^2 27^\circ + \cos^2 27^\circ}{\cos 27^\circ \cdot \sin 27^\circ} \\ &= \frac{1}{\sin 9^\circ \cdot \cos 9^\circ} - \frac{1}{\cos 27^\circ \cdot \sin 27^\circ} \\ &= \frac{2}{2 \sin 9^\circ \cos 9^\circ} - \frac{2}{2 \cos 27^\circ \sin 27^\circ} \\ &= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = \left(\frac{\sin 54^\circ - \sin 18^\circ}{\sin 18^\circ \cdot \sin 54^\circ} \right) \\ &= 2 \cdot \frac{2 \cos 36^\circ \cdot \sin 18^\circ}{\cos 36^\circ \cdot \sin 18^\circ} = 4 \end{aligned}$$

Question32

$\cos 6^\circ \sin 24^\circ \cos 72^\circ$ is equal to

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Options:

A. $-\frac{1}{8}$

B. $-\frac{1}{4}$

C. $\frac{1}{8}$

D. $\frac{1}{4}$

Answer: C

Solution:

$$\begin{aligned} & \cos 6^\circ \cdot \sin 24^\circ \cos 72^\circ \\ &= \frac{1}{2} \cdot 2 \cos 6^\circ \sin 24^\circ \cos 72^\circ \\ &= \frac{1}{2} (\sin 30^\circ + \sin 18^\circ) \cos 72^\circ \\ &= \frac{1}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{5}-1}{4} \right) \left(\frac{\sqrt{5}-1}{4} \right) \right] = \frac{1}{8} \end{aligned}$$

Question33

If $\sinh x = \frac{\sqrt{21}}{2}$, then $\cosh 2x + \sinh 2x$ is equal to

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Options:

A. $\frac{21}{2}$

B. $\frac{25}{2}$

C. $\frac{23+5\sqrt{21}}{2}$

D. $\frac{32+5\sqrt{23}}{2}$

Answer: C

Solution:

Given, $\sinh(x) = \frac{\sqrt{21}}{2}$

$$\Rightarrow \cosh x = \sqrt{1 + \sinh^2 x} = \sqrt{1 + \frac{21}{4}}$$

$$\Rightarrow \cosh x = \frac{5}{2}$$

Now, $\cosh 2x + \sinh 2x = 1 + 2 \sinh^2 x + 2 \sinh x \cosh x$

$$\begin{aligned} &= 1 + 2 \cdot \frac{21}{4} + 2 \times \frac{\sqrt{21}}{2} \times \frac{5}{2} \\ &= \frac{23}{2} + \sqrt{21} \times \frac{5}{2} = \frac{23 + 5\sqrt{21}}{2} \end{aligned}$$

Question34

If M_1 and M_2 are the maximum values of $\frac{1}{11 \cos 2x + 60 \sin 2x + 69}$ and $3 \cos^2 5x + 4 \sin^2 5x$ respectively, then $\frac{M_1}{M_2} =$

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Options:

A. $\frac{65}{2}$

B. $\frac{1}{32}$

C. $\frac{8}{3}$

D. 2

Answer: B

Solution:

$$\frac{1}{11 \cos 2x + 60 \sin 2x + 69}$$

denominator $11 \cos 2x + 60 \sin 2x + 69$

To find extrema of denominator

$$11 \cos 2x + 60 \sin 2x = \sqrt{11^2 + 60^2} \sin(2x + \theta)$$

$$\text{where, } \theta = \tan^{-1} \frac{11}{60}$$

$$= 61 \sin(2x + \theta)$$

This minimum value is -61 and the maximum value is 61 .

Therefore, $11 \cos 2x + 60 \sin 2x + 69$

$$= 69 - 61 = 8 \text{ to } 69 + 61 = 130$$

The maximum value is 130 and minimum is 8 of denominator

$$\text{Hence, } M_1 = \frac{1}{8}$$

For, $3 \cos^2 5x + 4 \sin^2 5x$

$$= 3(1 - \sin^2 5x) + 4 \sin^2 5x$$

$$= 3 + \sin^2 5x$$

The maximum value of $\sin^2 5x$ is 1 , hence

$$3 + \sin^2 5x = 3 + 1 = 4$$

because the range of $(\sin)^2$ is between 0 and 1

The minimum value of $\sin^2 5x$ is 0 .

Hence, $3 + \sin^2 5x \geq 3 + 0 = 3$

Thus, the maximum value $M_2 = 4$

Finally the ratio is

$$\frac{M_1}{M_2} = \frac{\frac{1}{8}}{4} = \frac{1}{8 \times 4} \Rightarrow \frac{M_1}{M_2} = \frac{1}{32}$$

Question35

$$4 \cos \frac{\pi}{7} \cos \frac{\pi}{5} \cos \frac{2\pi}{7} \cos \frac{2\pi}{5} \cos \frac{4\pi}{7} =$$

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Options:

A. $-\frac{1}{8}$

B. $\frac{1}{32}$

C. $-\frac{1}{32}$

D. $\frac{1}{8}$

Answer: A

Solution:

To solve the expression:

$$4 \cos \frac{\pi}{7} \cos \frac{\pi}{5} \cos \frac{2\pi}{7} \cos \frac{2\pi}{5} \cos \frac{4\pi}{7}$$

we can use known trigonometric identities and values. Direct computation might be cumbersome, so we leverage a known result for products of cosines.

Here is the important formula:

$$\cos \left(\frac{\pi}{7} \right) \cos \left(\frac{2\pi}{7} \right) \cos \left(\frac{3\pi}{7} \right) = \frac{1}{8}$$

Given the symmetry properties of cosine, specifically that:

$$\cos \left(\frac{4\pi}{7} \right) = -\cos \left(\frac{3\pi}{7} \right),$$

we can deduce:

$$\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = \frac{-1}{8}$$

Additionally, using the identity for the product of cosines:

$$\cos \frac{\pi}{5} \cos \frac{2\pi}{5} = \frac{1}{4}$$

Combining these results from equations (i) and (ii):

$$\begin{aligned} 4 \cos \frac{\pi}{7} \cos \frac{\pi}{5} \cos \frac{2\pi}{7} \cos \frac{2\pi}{5} \cos \frac{4\pi}{7} &= 4 \times \left(-\frac{1}{8}\right) \times \frac{1}{4} \\ &= \frac{-1}{8} \end{aligned}$$

Hence, the value of the expression is $-\frac{1}{8}$.

Question 36

If $\tanh x = \operatorname{sech} y = \frac{3}{5}$ and e^{x+y} is an integer, then $e^{x+y} =$

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Options:

A. 2

B. 8

C. 1

D. 6

Answer: D

Solution:

Step 1: Solve for $\tanh x = \frac{3}{5}$:

The hyperbolic tangent function is defined as:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{3}{5}$$

This implies:

$$\sinh x = \frac{3}{5} \cosh x$$

Let $\cosh x = z$, then:

$$\sinh x = \frac{3}{5} z$$

Using the identity:

$$\cosh^2 x - \sinh^2 x = 1$$

Substitute the values:

$$z^2 - \left(\frac{3}{5}z\right)^2 = 1$$

Solve this equation:

$$z^2 - \frac{9}{25}z^2 = 1$$

$$\frac{16}{25}z^2 = 1 \Rightarrow z^2 = \frac{25}{16}$$

Thus,

$$z = \frac{5}{4} \Rightarrow \cosh x = \frac{5}{4}, \sinh x = \frac{3}{4}$$

Step 2: Solve for $\operatorname{sech} y = \frac{3}{5}$:

The hyperbolic secant is defined as:

$$\operatorname{sech} y = \frac{1}{\cosh y} = \frac{3}{5}$$

Therefore:

$$\cosh y = \frac{5}{3}$$

Using the identity for hyperbolic functions:

$$\cosh^2 y - \sinh^2 y = 1$$

Calculate $\sinh y$:

$$\left(\frac{5}{3}\right)^2 - 1 = \sinh^2 y$$

$$\sinh^2 y = \frac{16}{9} \Rightarrow \sinh y = \frac{4}{3}$$

Step 3: Calculate e^{x+y} :

We need to find $e^x \cdot e^y$.

For e^x :

$$e^x = \cosh x + \sinh x = \frac{5}{4} + \frac{3}{4} = 2$$

For e^y :

$$e^y = \cosh y + \sinh y = \frac{5}{3} + \frac{4}{3} = 3$$

Thus:

$$e^{x+y} = e^x \cdot e^y = 2 \cdot 3 = 6$$

Therefore, the value of e^{x+y} is 6.

Question 37

If A, B, C are the angles of triangle, then $\sin 2A - \sin 2B + \sin 2C =$

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Options:

A. $4 \cos A \cos B \sin C$

B. $4 \cos A \sin B \cos C$

C. $4 \cos A \sin B \cos C - 1$

D. $4 \sin A \cos B \sin C$

Answer: B

Solution:

Given that A, B and C are the angle of triangle

$$\begin{aligned} A + B + C &= \pi \\ 2A + 2B + 2C &= 2\pi \quad \dots (i) \end{aligned}$$

Now, $\sin 2A - \sin 2B + \sin 2C$

$$\begin{aligned} &= \sin 2A + \sin 2C - \sin 2B \\ &= 2 \sin \left(\frac{2A + 2C}{2} \right) \cos \left(\frac{2A - 2C}{2} \right) - \sin[2\pi - 2(A + C)] \end{aligned}$$

[from Eq. (i)]

$$\begin{aligned} &= 2 \sin(A + C) \cos(A - C) + \sin 2(A + C) \\ &= 2 \sin(A + C) \cos(A - C) + 2 \sin(A + C) \cos(A + C) \\ &= 2 \sin(A + C) [\cos(A - C) + \cos(A + C)] \\ &= 2 \sin(\pi - B) [2 \cos A \cos C] \\ &= 4 \cos A \sin B \cos C \end{aligned}$$

Question38

Assertion (A) : If $A = 10^\circ, B = 16^\circ$ and $C = 19^\circ$, then $\tan 2A \tan 2B + \tan 2B \tan 2C + \tan 2C \tan 2A = 1$

Reason (R) : If $A + B + C = 180^\circ, \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}$

$$= \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}$$

Which of the following is correct ?

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Options:

- A. Both (A) and (R) are true and (R) is the correct explanation of (A)
- B. Both (A) and (R) are true and (R) is not correct explanation of (A)
- C. (A) is true, (R) is false
- D. (A) is false, (R) is true.

Answer: A

Solution:

In case of assertion, if $A = 10^\circ$,

$$B = 16^\circ \text{ and } C = 19^\circ$$

$$\therefore A + B + C = 180^\circ$$

$$\Rightarrow 2A + 2B + 2C = \frac{\pi}{2}$$

$$\Rightarrow 2A + 2B = \frac{\pi}{2} - 2C$$

On taking $\tan$ both sides, we get

$$\tan(2A + 2B) = \tan\left(\frac{\pi}{2} - 2C\right)$$

$$\Rightarrow \frac{\tan 2A + \tan 2B}{1 - \tan 2A \tan 2B} = \frac{1}{\tan 2C}$$

$$\Rightarrow \tan 2A \tan 2C + \tan 2B \tan 2C = 1 - \tan 2A \tan 2B$$

$$\tan 2A \tan 2C + \tan 2B \tan 2C + \tan 2A \tan 2B = 1$$

So, assertion is true.

In case of reason,

$$\text{Since, } A + B + C = 180^\circ$$

$$\frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$$

$$\Rightarrow \frac{A}{2} + \frac{B}{2} = \frac{\pi}{2} - \frac{C}{2}$$

$$\Rightarrow \cot\left(\frac{A}{2} + \frac{B}{2}\right) = \cot\left(\frac{\pi}{2} - \frac{C}{2}\right)$$

$$\Rightarrow \frac{\cot \frac{A}{2} \cot \frac{B}{2} - 1}{\cot \frac{A}{2} + \cot \frac{B}{2}} = \tan \frac{C}{2}$$

$$\Rightarrow \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2} = \cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}$$

So, reason is true.

Question39

If α is in the 3rd quadrant, β is in the 2nd quadrant such that

$$\tan \alpha = \frac{1}{7}, \sin \beta = \frac{1}{\sqrt{10}}, \text{ then } \sin(2\alpha + \beta) =$$

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Options:

A. $\frac{3 \times \sqrt{10}}{25}$

B. $\frac{3}{\sqrt{10}}$

C. $\frac{3}{25\sqrt{10}}$

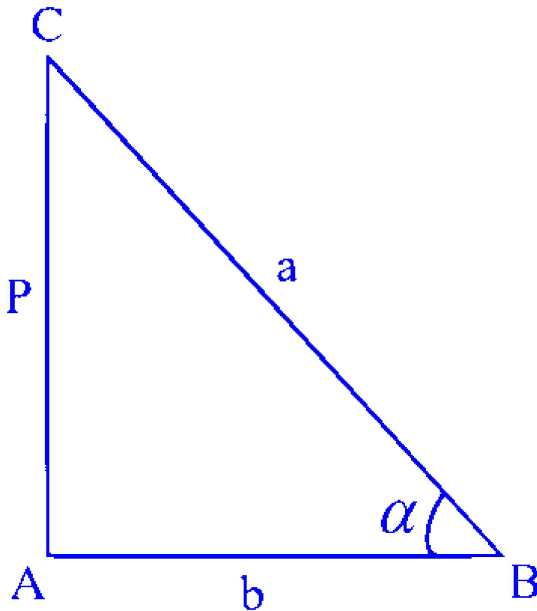
D. $\frac{\sqrt{10}}{3 \times 25}$

Answer: C

Solution:

$$\text{Given, } \tan \alpha = \frac{1}{7}, \sin \beta = \frac{1}{\sqrt{10}}$$

$$\tan \alpha = \frac{p}{b} = \frac{1}{7}$$



Using Pythagoras theorem,

$$\begin{aligned} p^2 + b^2 &= a^2 \Rightarrow a = \sqrt{1^2 + 7^2} = \sqrt{50} \\ \sin \alpha &= \frac{1}{\sqrt{50}} \text{ and } \cos \alpha = \frac{7}{\sqrt{50}} \\ \text{and } \sin \beta &= \frac{1}{\sqrt{10}} \\ \cos^2 \beta &= 1 - \left(\frac{1}{\sqrt{10}} \right)^2 \quad [\because \sin^2 \alpha + \cos^2 \alpha = 1] \\ \Rightarrow \cos^2 \beta &= \frac{9}{10} \Rightarrow \cos \beta = -\frac{3}{\sqrt{10}} \end{aligned}$$

[$\because \beta$ is in 3rd quadrant]

$$\text{Now, } \sin(2\alpha + \beta) = \sin 2\alpha \cos \beta + \cos 2\alpha \sin \beta$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha = 2 \cdot \frac{1}{\sqrt{50}} \cdot \frac{7}{\sqrt{50}} = \frac{14}{50}$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = \frac{49}{50} - \frac{1}{50} = \frac{48}{50}$$

On putting all these values in Eq. (i), we get

$$\begin{aligned} \sin(2\alpha + \beta) &= \frac{14}{50} \cdot \left(-\frac{3}{\sqrt{10}} \right) + \frac{48}{50} \cdot \frac{1}{\sqrt{10}} \\ &= -\frac{42}{50\sqrt{10}} + \frac{48}{50\sqrt{10}} = \frac{1}{50\sqrt{10}} (48 - 42) \\ &= \frac{3}{25\sqrt{10}} \end{aligned}$$

Question40

If the period of the function $f(x) = \frac{\tan 5x \cos 3x}{\sin 6x}$ is α , then $f\left(\frac{\alpha}{8}\right) =$

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Options:

A. $\frac{1}{2}$

B. -1

C. $\frac{1}{\sqrt{2}}$

D. $-\frac{1}{\sqrt{2}}$

Answer: C

Solution:

Let $f(x) = \frac{\tan(5x) \cos 3x}{\sin 6x}$

1.Period of $\tan(5x) = \frac{\pi}{5}$

2.Period of $\cos 3x = \frac{2\pi}{3}$

3.Period of $\sin 6x = \frac{\pi}{3}$

The period of $f(x)$ is the least common multiple of $\frac{\pi}{5}$, $\frac{2\pi}{3}$ and $\frac{\pi}{3}$.

$\text{LCM}\left(\frac{\pi}{3}, \frac{2\pi}{3}, \frac{\pi}{3}\right) = 2\pi$

$\therefore \alpha = 2\pi$

$$\begin{aligned} \text{Now, } f\left(\frac{\alpha}{8}\right) &= f\left(\frac{\pi}{4}\right) = \frac{\tan\left(\frac{5\pi}{4}\right) \cos \frac{3\pi}{4}}{\sin \frac{3\pi}{2}} \\ &= \frac{\tan \frac{\pi}{4} \cdot (-\cos \frac{\pi}{4})}{-\sin \frac{\pi}{2}} \\ &= \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} \end{aligned}$$

Question41

If $\sin x + \sin y = \alpha$, $\cos x + \cos y = \beta$, then $\operatorname{cosec}(x + y) =$

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Options:

A. $\frac{\beta^2 - \alpha^2}{\beta^2 + \alpha^2}$

B. $\frac{2\beta\alpha}{\beta^2 - \alpha^2}$

C. $\frac{\beta^2 + \alpha^2}{2\beta\alpha}$

D. $\frac{2a\beta}{\beta^2 + \alpha^2}$

Answer: C

Solution:

If $\sin x + \sin y = \alpha$

$\cos x + \cos y = \beta$

$\operatorname{cosec}(x + y) = ?$

From Eq. (i), we get

$$2 \sin \left(\frac{x+y}{2} \right) \cdot \cos \left(\frac{x-y}{2} \right) = \alpha$$

$$\left(\because \sin C + \sin D = 2 \sin \frac{(C+D)}{2} \cos \frac{(C-D)}{2} \right)$$

and using Eq. (ii)

$$2 \cos \left(\frac{x-y}{2} \right) \cdot \cos \left(\frac{x+y}{2} \right) = \beta$$

$$\left(\because \cos C + \cos D = 2 \cos \left(\frac{C-D}{2} \right) \cdot \cos \left(\frac{C+D}{2} \right) \right)$$

On dividing Eq. (iii) by Eq. (iv), we get

$$\tan \left(\frac{x+y}{2} \right) = \frac{\alpha}{\beta}$$

$$\left(\because \tan^2 \theta = \sec^2 \theta - 1 \text{ and } \sec \theta = \frac{1}{\cos \theta} \right)$$

So, $\cos \left(\frac{x+y}{2} \right) = \left[\frac{\beta^2}{\alpha^2 + \beta^2} \right]^{\frac{1}{2}}$

$$\sin \left(\frac{x+y}{2} \right) = \left(1 - \cos^2 \left(\frac{x+y}{2} \right) \right)^{\frac{1}{2}}$$

$$= \left[\frac{\alpha^2}{\alpha^2 + \beta^2} \right]^{\frac{1}{2}}$$

$$\sin(x+y) = 2 \sin \left(\frac{x+y}{2} \right) \cdot \cos \left(\frac{x+y}{2} \right)$$

$$\left(\because \sin A = \frac{2 \sin \frac{A}{2} \cos \frac{A}{2}}{2} \right)$$

From Eqs. (vi) and (vii), we get

$$\begin{aligned}\sin(x+y) &= \frac{2\alpha}{(\alpha^2 + \beta^2)^{\frac{1}{2}}} \cdot \frac{\beta}{(\alpha^2 + \beta^2)^{\frac{1}{2}}} \\ &= \frac{2\alpha\beta}{(\alpha^2 + \beta^2)} \left(\because \operatorname{cosec} \theta = \frac{1}{\sin \theta} \right) \\ \operatorname{cosec}(x+y) &= \frac{1}{\sin(x+y)} \\ &= \frac{1}{\frac{2\alpha\beta}{(\alpha^2 + \beta^2)}} = \frac{\alpha^2 + \beta^2}{2\alpha\beta}\end{aligned}$$

Question 42

If $P + Q + R = \frac{\pi}{4}$, then $\cos\left(\frac{\pi}{8} - P\right) + \cos\left(\frac{\pi}{8} - Q\right) + \cos\left(\frac{\pi}{8} - R\right) =$

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Options:

- A. $4 \cos \frac{P}{2} \cos \frac{Q}{2}, \cos \frac{R}{2} - \cos \frac{\pi}{8}$
- B. $4 \cos \frac{P}{2} \cos \frac{Q}{2} \cdot \sin \frac{R}{2} + \cos \frac{\pi}{8}$
- C. $4 \sin \frac{P}{2} \cos \frac{Q}{2}, \sin \frac{R}{2} - \cos \frac{\pi}{8}$
- D. $4 \sin \frac{P}{2} \cos \frac{Q}{2}, \sin \frac{R}{2} - \cos \frac{\pi}{8}$

Answer: A

Solution:

Given, $P + Q + R = \frac{\pi}{4}$

$$\therefore \frac{P}{2} + \frac{Q}{2} + \frac{R}{2} = \frac{\pi}{8}$$

Now, $\cos\left(\frac{\pi}{8} - P\right) + \cos\left(\frac{\pi}{8} - Q\right) + \cos\left(\frac{\pi}{8} - R\right)$

$$\begin{aligned}
&= 2 \cos \left(\frac{\left(\frac{\pi}{8} - P \right) + \left(\frac{\pi}{8} - Q \right)}{2} \right) \\
&\cos \left(\frac{\left(\frac{\pi}{8} - P \right) - \left(\frac{\pi}{8} - Q \right)}{2} \right) + \cos \left(\frac{\pi}{8} - R \right) \\
&= 2 \cos \left(\frac{\pi}{8} - \frac{P}{2} - \frac{Q}{2} \right) \cos \left(\frac{Q}{2} - \frac{P}{2} \right) + \cos \left(\frac{\pi}{8} - R \right) \\
&= 2 \cos \left(\frac{R}{2} \right) \cos \left(\frac{Q}{2} - \frac{P}{2} \right) + \cos \left(\frac{P}{2} + \frac{Q}{2} - \frac{R}{2} \right) \\
&= 2 \cos \left(\frac{R}{2} \right) \cos \left(\frac{Q}{2} - \frac{P}{2} \right) + \cos \left(\frac{P}{2} + \frac{Q}{2} \right) \\
&= 2 \cos \left(\frac{R}{2} \right) + \sin \left(\frac{P}{2} + \frac{Q}{2} \right) \sin \left(\frac{R}{2} \right) \\
&- \cos \left(\frac{Q}{2} - \frac{P}{2} \right) + \cos \left(\frac{P}{2} + \frac{Q}{2} \right) \Big] \\
&= 2 \cos \left(\frac{R}{2} + \frac{Q}{2} \right) \cos \left(\frac{R}{2} \right) + \sin \left(\frac{P}{2} + \frac{Q}{2} \right) \sin \left(\frac{R}{2} \right) \\
&- \left[\cos \left(\frac{P}{2} + \frac{P}{2} + \frac{Q}{2} \right) \cos \left(\frac{R}{2} \right) - \cos \left(\frac{Q}{2} - \frac{P}{2} \right) \right] \\
&= 2 \cos \left(\frac{R}{2} \right) \left[2 \cos \left(\frac{Q}{2} \right) \cos \left(\frac{P}{2} \right) \right] - \cos \left(\frac{P}{2} + \frac{Q}{2} + \frac{R}{2} \right) \\
&= 4 \cos \left(\frac{P}{2} \right) \cos \left(\frac{Q}{2} \right) \cos \left(\frac{R}{2} \right) - \cos \left(\frac{\pi}{8} \right)
\end{aligned}$$

Question43

If θ is an acute angle, $\cosh x = K$ and $\sinh x = \tan \theta$, then $\sin \theta =$

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Options:

- A. $\frac{k}{k^2+1}$
- B. $\frac{k^2+1}{k^2+2}$
- C. $\frac{\sqrt{k^2-1}}{k}$
- D. $\frac{\sqrt{k^2-1}}{\sqrt{k^2+1}}$

Answer: C

Solution:

Given that θ is an acute angle, $\cosh x = k$, and $\sinh x = \tan \theta$, we can establish the following relationships:

First, note that:

$$\tan \theta = \sinh x$$

Squaring both sides, we obtain:

$$\tan^2 \theta = \sinh^2 x$$

We also have the trigonometric identity:

$$\sec^2 \theta = 1 + \tan^2 \theta$$

Substituting $\sinh^2 x$ for $\tan^2 \theta$ gives:

$$\sec^2 \theta = 1 + \sinh^2 x$$

Next, recall the identity for hyperbolic functions:

$$\cosh^2 x - \sinh^2 x = 1$$

This implies:

$$\cosh^2 x = 1 + \sinh^2 x$$

Given $\cosh x = k$, it follows that

$$\cosh^2 x = k^2$$

Thus:

$$\sec^2 \theta = \cosh^2 x = k^2$$

Taking the reciprocal to find $\cos^2 \theta$:

$$\cos^2 \theta = \frac{1}{k^2}$$

Using the Pythagorean identity:

$$1 = \sin^2 \theta + \cos^2 \theta$$

Substitute the expression for $\cos^2 \theta$:

$$1 = \sin^2 \theta + \frac{1}{k^2}$$

Rearranging for $\sin^2 \theta$ gives:

$$\sin^2 \theta = 1 - \frac{1}{k^2}$$

This simplifies to:

$$\sin^2 \theta = \frac{k^2 - 1}{k^2}$$

Taking the square root, given that θ is an acute angle (which ensures $\sin \theta$ is positive), we find:

$$\sin \theta = \frac{\sqrt{k^2 - 1}}{k}$$

Question44

If $\sec \theta + \tan \theta = \frac{1}{3}$, then the quadrant in which 2θ lies is

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Options:

A. 1st quadrant

B. 2nd quadrant

C. 3rd quadrant

D. 4th quadrant

Answer: C

Solution:

Given:

$$\sec \theta + \tan \theta = \frac{1}{3} \quad \dots (i)$$

We know the identity:

$$\sec^2 \theta - \tan^2 \theta = 1$$

This can be rewritten as:

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

Substituting the given value from equation (i):

$$\frac{1}{3}(\sec \theta - \tan \theta) = 1$$

Solving for $\sec \theta - \tan \theta$:

$$\sec \theta - \tan \theta = 3 \quad \dots (ii)$$

Using equations (i) and (ii), we solve for $\sec \theta$ and $\tan \theta$:

$$\sec \theta = \frac{5}{3}$$

$$\tan \theta = \frac{-4}{3}$$

Now, we find $\sin \theta$:

$$\sin \theta = \frac{\tan \theta}{\sec \theta} = \frac{\frac{-4}{3}}{\frac{5}{3}} = \frac{-4}{5}$$

Here, both $\sin \theta$ and $\tan \theta$ are negative, indicating that θ lies in the IVth quadrant.

Calculating $\sin 2\theta$:

$$\sin 2\theta = 2 \sin \theta \cos \theta = 2 \times \frac{-4}{5} \times \frac{3}{5} = \frac{-24}{25}$$

Next, calculating $\tan 2\theta$:

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{24}{7}$$

Since $\sin 2\theta$ is negative and $\tan 2\theta$ is positive, 2θ lies in the IIIrd quadrant.

Question45

If $540^\circ < A < 630^\circ$ and $|\cos A| = \frac{5}{13}$, then $\tan \frac{A}{2} \tan A =$

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Options:

A. $\frac{18}{5}$

B. $\frac{8}{5}$

C. $-\frac{8}{5}$

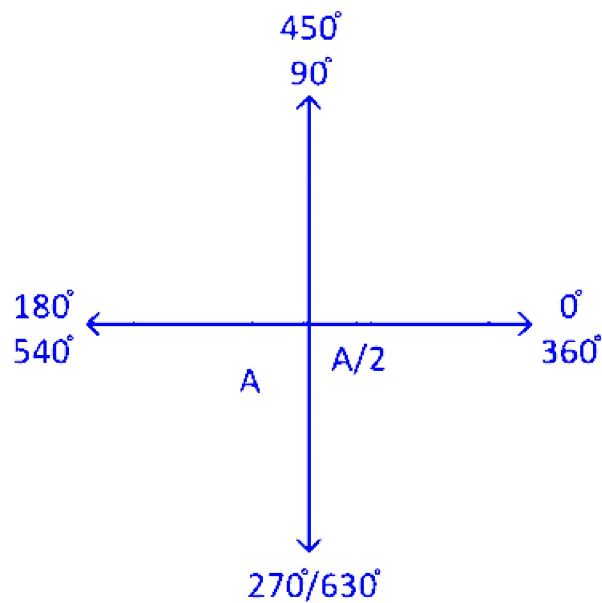
D. $-\frac{18}{5}$

Answer: D

Solution:

Since, $540^\circ < A < 630^\circ$

$$\therefore 270^\circ < \frac{A}{2} < 315^\circ$$



Since, A lies in IIIrd quadrant $\therefore \tan A$ is positive and $\frac{A}{2}$ lies in IVth quadrant $\therefore \tan \frac{A}{2}$ is negative.

Now, $\cos A = \frac{5}{13}$

$$\Rightarrow \tan A = \frac{12}{5}$$

Also, $\tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}$

$$\Rightarrow \frac{12}{5} - \frac{12}{5} \tan^2 \frac{A}{2} = 2 \tan \frac{A}{2}$$

$$\Rightarrow 6 \tan^2 \frac{A}{2} + 5 \tan \frac{A}{2} - 6 = 0$$

$$\Rightarrow \tan \frac{A}{2} = \frac{-3}{2}$$

$$\therefore \tan \frac{A}{2} \tan A = \frac{-18}{5}$$

Question46

If $(\alpha + \beta)$ is not a multiple of $\frac{\pi}{2}$ and $3 \sin(\alpha - \beta) = 5 \cos(\alpha + \beta)$, then $\tan\left(\frac{\pi}{4} + \alpha\right) + 4 \tan\left(\frac{\pi}{4} + \beta\right) =$

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Options:

A. 0

B. 1

C. 4

D. 2

Answer: A

Solution:

Given, $3 \sin(\alpha - \beta) = 5 \cos(\alpha + \beta)$

$$\frac{\sin(\alpha - \beta)}{\cos(\alpha + \beta)} = \frac{5}{3}$$
$$\frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta} = \frac{5}{3}$$

Divide both numerator and denominator by $\cos \alpha \cos \beta$

$$\frac{\tan \alpha - \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{5}{3}$$
$$\Rightarrow 3 \tan \alpha - 3 \tan \beta = 5 - 5 \tan \alpha \tan \beta$$
$$\Rightarrow 3 \tan \alpha - 3 \tan \beta - 5 + 5 \tan \alpha \tan \beta = 0$$

Now, $\tan\left(\frac{\pi}{4} + \alpha\right) + 4 \tan\left(\frac{\pi}{4} + \beta\right)$

$$= \frac{1 + \tan \alpha}{1 - \tan \alpha} + 4 \left(\frac{1 + \tan \beta}{1 - \tan \beta} \right)$$
$$= \frac{(1 - \tan \beta + \tan \alpha - \tan \alpha \tan \beta + 4)}{(1 - \tan \alpha)(1 - \tan \beta)}$$
$$= \frac{3 \tan \beta - 3 \tan \alpha + 5 - 5 \tan \alpha \tan \beta}{(1 - \tan \alpha)(1 - \tan \beta)}$$
$$= \frac{-(3 \tan \alpha - 3 \tan \beta - 5 + 5 \tan \alpha \tan \beta)}{(1 - \tan \alpha)(1 - \tan \beta)}$$
$$= 0$$

Question 47

If $\cos \alpha + \cos \beta + \cos \gamma = \sin \alpha + \sin \beta + \sin \gamma = 0$, then $(\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma)^2 + (\sin^3 \alpha + \sin^3 \beta + \sin^3 \gamma)^2 =$

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Options:

A. 1

B. $\frac{3}{4}$ C. $\frac{9}{16}$ D. $\frac{9}{8}$ **Answer: C****Solution:**

We have,

 $\cos \alpha + \cos \beta + \cos \gamma = \sin \alpha + \sin \beta + \sin \gamma = 0$ then,

$$\begin{aligned}
 & (\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma)^2 + \\
 & (\sin^3 \alpha + \sin^3 \beta + \sin^3 \gamma)^2 \\
 &= (3 \cos \alpha \cos \beta \cos \gamma)^2 + (3 \sin \alpha \sin \beta \sin \gamma)^2 \\
 & \left[\begin{array}{l} \because \text{ If } x + y + z = 0 \\ \therefore x^3 + y^3 + z^3 = 3xyz \end{array} \right] \\
 &= 9(\cos \alpha \cos \beta \cos \gamma)^2 + 9(\sin \alpha \sin \beta \sin \gamma)^2
 \end{aligned}$$

So, let $\alpha = \theta, \beta = \theta + \frac{2\pi}{3}$ and $\gamma = \theta + \frac{4\pi}{3}$ then, $\cos \alpha \cos \beta \cos \gamma = \cos \theta \cos \left(\theta + \frac{2\pi}{3} \right)$

$$\begin{aligned}
 & \cos \left(\theta + \frac{4\pi}{3} \right) \\
 &= \cos \theta \left[\cos \theta \cos \frac{2\pi}{3} - \sin \theta \sin \frac{2\pi}{3} \right] \\
 & \left[\cos \theta \cos \frac{4\pi}{3} - \sin \theta \sin \frac{4\pi}{3} \right] \\
 &= \cos \theta \left(-\frac{1}{2} \cos \theta - \frac{\sqrt{3}}{2} \sin \theta \right) \\
 &= \cos \theta \left(\frac{1}{4} \cos^2 \theta - \frac{3}{4} \sin^2 \theta \right) \\
 &= \frac{1}{4} \cos^3 \theta - \frac{1}{4} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta \sin^2 \theta
 \end{aligned}$$

Similarly,

$$\sin \alpha \sin \beta \sin \gamma = \frac{3}{4} \sin \theta \cos^2 \theta - \frac{1}{4} \sin^3 \theta$$

From Eqs. (ii) and (iii) in Eq. (i), we get

$$\begin{aligned}
 & (\cos^3 \alpha + \cos^3 \beta + \cos^3 \gamma)^2 \\
 & + (\sin^3 \alpha + \sin^3 \beta + \sin^3 \gamma)^2
 \end{aligned}$$

$$\begin{aligned}
&= \frac{9\left(\frac{1}{4}\cos^3\theta - \frac{3}{4}\cos\theta\sin^2\theta\right)^2 + 9}{\left(\frac{3}{4}\sin\theta\cos^2\theta - \frac{1}{4}\sin^3\theta\right)^2} \\
&= \frac{9}{16} \left[(\cos^3\theta - 3\cos\theta\sin^2\theta)^2 \right. \\
&\quad \left. + (3\sin\theta\cos^2\theta - \sin^3\theta)^2 \right] \\
&= \frac{9}{16} [\cos^6\theta + 9\cos^2\theta\sin^4\theta \\
&\quad - 6\cos^4\theta\sin^2\theta + 9\sin^2\theta\cos^4\theta \\
&= \frac{9}{16} [\cos^6\theta + \sin^6\theta - 6\cos^2\theta\sin^4\theta] \\
&= \frac{9}{16} [(\cos^2\theta)^3\sin^4\theta \\
&= (\sin^2\theta)^3 + 3\cos^2\theta \\
&= \frac{9}{16} (\cos^2\theta + \sin^2\theta)^3 \\
&= \frac{9}{16} \times (1)^3 = \frac{9}{16}
\end{aligned}$$

Question48

$$\frac{\cos 10^\circ + \cos 80^\circ}{\sin 80^\circ - \sin 10^\circ} =$$

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Options:

A. $\tan 35^\circ$

B. $\tan 55^\circ$

C. $\tan 20^\circ$

D. $\tan 70^\circ$

Answer: B

Solution:

To solve the given expression:

$$\frac{\cos 10^\circ + \cos 80^\circ}{\sin 80^\circ - \sin 10^\circ}$$

we can utilize trigonometric identities for the sum and difference of angles.

Start with the identity for the sum of cosines:

$$\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

For $\cos 10^\circ + \cos 80^\circ$, set $A = 10^\circ$ and $B = 80^\circ$:

$$\cos 10^\circ + \cos 80^\circ = 2 \cos \left(\frac{10^\circ + 80^\circ}{2} \right) \cos \left(\frac{10^\circ - 80^\circ}{2} \right) = 2 \cos 45^\circ \cos(-35^\circ)$$

For the difference of sines, use the identity:

$$\sin A - \sin B = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

For $\sin 80^\circ - \sin 10^\circ$, set $A = 80^\circ$ and $B = 10^\circ$:

$$\sin 80^\circ - \sin 10^\circ = 2 \cos \left(\frac{80^\circ + 10^\circ}{2} \right) \sin \left(\frac{80^\circ - 10^\circ}{2} \right) = 2 \cos 45^\circ \sin 35^\circ$$

Substitute these results back into the original expression:

$$\frac{2 \cos 45^\circ \cos(-35^\circ)}{2 \cos 45^\circ \sin 35^\circ}$$

Cancel the common factor $2 \cos 45^\circ$:

$$\frac{\cos 35^\circ}{\sin 35^\circ} = \cot 35^\circ = \tan(90^\circ - 35^\circ) = \tan 55^\circ$$

Therefore, the expression evaluates to $\tan 55^\circ$.

Question49

$$\frac{\sin 1^\circ + \sin 2^\circ + \dots + \sin 89^\circ}{2(\cos 1^\circ + \cos 2^\circ + \dots + \cos 44^\circ) + 1} =$$

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Options:

A. $\sqrt{2}$

B. $\frac{1}{\sqrt{2}}$

C. 2

D. $\frac{1}{2}$

Answer: B

Solution:

To solve the given problem, we start by simplifying the expression:

$$\frac{\sin 1^\circ + \sin 2^\circ + \dots + \sin 89^\circ}{2(\cos 1^\circ + \cos 2^\circ + \dots + \cos 44^\circ) + 1}$$

First, we recognize that the sum in the numerator can be grouped as follows:

$$(\sin 89^\circ + \sin 1^\circ) + (\sin 88^\circ + \sin 2^\circ) + \dots + \sin 45^\circ$$

Utilizing the identity $\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$, each paired term becomes:

$$2 \sin \left(\frac{89+1}{2} \right) \cos \left(\frac{89-1}{2} \right) + \dots + 2 \sin \left(\frac{46+44}{2} \right) \cos \left(\frac{46-44}{2} \right) + \sin 45^\circ$$

This can be written as:

$$2(\sin 45^\circ \cos 44^\circ + \sin 45^\circ \cos 43^\circ + \dots + \sin 45^\circ \cos 1^\circ) + \sin 45^\circ$$

The denominator can also be broken down nicely:

$$2(\cos 1^\circ + \cos 2^\circ + \dots + \cos 44^\circ) + 1$$

Now, extracting $\sin 45^\circ$ from the numerator terms gives us:

$$2 \sin 45^\circ (\cos 44^\circ + \cos 43^\circ + \dots + \cos 1^\circ + 1)$$

The entire expression simplifies to:

$$\frac{2 \sin 45^\circ (\cos 44^\circ + \cos 43^\circ + \dots + \cos 1^\circ + 1)}{2(\cos 1^\circ + \cos 2^\circ + \dots + \cos 44^\circ) + 1}$$

Since $\sin 45^\circ = \frac{1}{\sqrt{2}}$, the expression becomes:

$$\frac{\sin 45^\circ}{\cos 45^\circ} = \frac{1}{\sqrt{2}}$$

Question 50

The value of $5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3$ lies between

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Options:

A. -2 and 5

B. -1 and 8

C. -3 and 6

D. -4 and 10

Answer: D

Solution:

Let's analyze the expression $5 \cos \theta + 3 \cos \left(\theta + \frac{\pi}{3} \right) + 3$.

First, we expand the cosine term using the cosine addition formula:

$$\cos \left(\theta + \frac{\pi}{3} \right) = \cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3}.$$

Substitute this back into the original expression:

$$5 \cos \theta + 3 \left(\cos \theta \cos \frac{\pi}{3} - \sin \theta \sin \frac{\pi}{3} \right) + 3.$$

Simplify by plugging in the values for $\cos \frac{\pi}{3} = \frac{1}{2}$ and $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$:

$$= 5 \cos \theta + \frac{3}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta + 3.$$

$$= \frac{1}{2} (13 \cos \theta - 3\sqrt{3} \sin \theta) + 3.$$

Now, let's express this in the form $r \cos(\alpha + \theta)$ such that:

$$13 = r \cos \alpha \quad \text{and} \quad 3\sqrt{3} = r \sin \alpha.$$

We find r by:

$$r = \sqrt{13^2 + (3\sqrt{3})^2} = \sqrt{169 + 27} = \sqrt{196} = 14.$$

Substitute r back into:

$$= \frac{1}{2}[r \cos(\alpha + \theta)] + 3 = \frac{1}{2}[14 \cos(\alpha + \theta)] + 3 = 7 \cos(\alpha + \theta) + 3.$$

The range of the cosine function is $-1 \leq \cos(\alpha + \theta) \leq 1$. Therefore:

$$-7 + 3 \leq 7 \cos(\alpha + \theta) + 3 \leq 7 + 3.$$

$$-4 \leq 7 \cos(\alpha + \theta) + 3 \leq 10.$$

Thus, the value of the expression lies within the interval $[-4, 10]$.

Question 51

Statement (S1) $\sin 55^\circ + \sin 53^\circ - \sin 19^\circ - \sin 17^\circ = \cos 2^\circ$

Statement (S2) Range of $\frac{1}{3 - \cos 2x}$ is $\left[\frac{1}{4}, \frac{1}{2}\right]$

Which one of the following is correct?

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Options:

- A. Both (S1) and (S2) are true.
- B. Both (S 1) and (S 2) are false.
- C. (S1) is true, (S2) is false.
- D. (S1) is false, (S2) is true.

Answer: D

Solution:

Clarification of Statements S1 and S2

Statement S1:

We need to verify the equation:

$$\sin 55^\circ + \sin 53^\circ - \sin 19^\circ - \sin 17^\circ = \cos 2^\circ$$

Derivation:

Use the sum-to-product identities:

$$\sin 55^\circ + \sin 53^\circ = 2 \sin 54^\circ \cos 1^\circ$$

$$\sin 19^\circ + \sin 17^\circ = 2 \sin 18^\circ \cos 1^\circ$$

Subtract the two results:

$$2 \sin 54^\circ \cos 1^\circ - 2 \sin 18^\circ \cos 1^\circ = 2 \cos 1^\circ (\sin 54^\circ - \sin 18^\circ)$$

Apply the difference of sines identity:

$$\sin A - \sin B = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$\sin 54^\circ - \sin 18^\circ = \cos 36^\circ - \sin 18^\circ$$

Use known trigonometric values:

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4}, \quad \sin 18^\circ = \frac{\sqrt{5}-1}{4}$$

$$\cos 36^\circ - \sin 18^\circ = \frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} = \frac{1}{2}$$

Substitute back:

$$2 \cos 1^\circ \frac{1}{2} = \cos 1^\circ$$

Therefore, S1 is false because the left expression simplifies to $\cos 1^\circ$, not $\cos 2^\circ$.

Statement S2:

We need to find the range of:

$$\frac{1}{3 - \cos 2x}$$

Derivation:

The range of $\cos 2x$ is $[-1, 1]$.

Translate the range:

$$3 - \cos 2x \Rightarrow [2, 4]$$

The range of the reciprocal:

$$\frac{1}{3 - \cos 2x} \Rightarrow \left[\frac{1}{4}, \frac{1}{2} \right]$$

Therefore, S2 is true.

In conclusion, Statement S1 is false, and Statement S2 is true.

Question 52

$$\tan 6^\circ + \tan 42^\circ + \tan 66^\circ + \tan 78^\circ =$$

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Options:

A. 0

B. 1

C. 2

D. 3

Answer: B

Solution:

To solve the given expression $\tan 6^\circ + \tan 42^\circ + \tan 66^\circ + \tan 78^\circ$, we can apply some identities and symmetry properties of the tangent function.

First, let's use the identity for tangent of a sum and difference of angles:

$$\tan(A + B) \tan(A - B) = \left(\frac{\tan A + \tan B}{1 - \tan A \tan B} \right) \left(\frac{\tan A - \tan B}{1 + \tan A \tan B} \right)$$

This can be simplified to:

$$\tan(A + B) \tan(A - B) = \frac{\tan^2 A - \tan^2 B}{1 - \tan^2 A \tan^2 B} \quad \dots (i)$$

Now, substitute $A = 60^\circ$ and $B = 18^\circ$ into equation (i):

$$\tan 78^\circ \times \tan 42^\circ = \frac{\tan^2 60^\circ - \tan^2 18^\circ}{1 - \tan^2 60^\circ \tan^2 18^\circ} = \frac{3 - \tan^2 18^\circ}{1 - 3 \tan^2 18^\circ}$$

Simplifying:

$$= \frac{1}{\tan 18^\circ} \left[\frac{3 \tan 18^\circ - \tan^3 18^\circ}{1 - 3 \tan^2 18^\circ} \right]$$

So,

$$\tan 78^\circ \tan 42^\circ = \frac{\tan 54^\circ}{\tan 18^\circ} \quad \dots (ii)$$

Next, substitute $A = 60^\circ$ and $B = 54^\circ$ in equation (i):

$$\tan 114^\circ \times \tan 6^\circ = \frac{3 - \tan^2 54^\circ}{1 - 3 \tan^2 54^\circ} = \frac{1}{\tan 54^\circ} \left[\frac{3 \tan 54^\circ - \tan^3 54^\circ}{1 - 3 \tan^2 54^\circ} \right]$$

This gives:

$$= \frac{\tan 162^\circ}{\tan 54^\circ}$$

Since $\tan 114^\circ = \tan(180^\circ - 66^\circ) = -\tan 66^\circ$ and $\tan 162^\circ = \tan(180^\circ - 18^\circ) = -\tan 18^\circ$, we have:

$$\tan 66^\circ \times \tan 6^\circ = \frac{\tan 18^\circ}{\tan 54^\circ} \quad \dots (iii)$$

Multiplying equations (ii) and (iii):

$$\tan 6^\circ \tan 42^\circ \tan 66^\circ \tan 78^\circ = \frac{\tan 54^\circ}{\tan 18^\circ} \times \frac{\tan 18^\circ}{\tan 54^\circ} = 1$$

Therefore, $\tan 6^\circ + \tan 42^\circ + \tan 66^\circ + \tan 78^\circ$ sums up to a neat result based on symmetrical properties of angles and tangent function identities.

Question53

The maximum value of $12 \sin x - 5 \cos x + 3$ is

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Options:

- A. 18
- B. 13
- C. 16
- D. 10

Answer: C

Solution:

To find the maximum value of the expression $12 \sin x - 5 \cos x + 3$, we start by considering the form $a \sin x + b \cos x$, which achieves its maximum value at $\sqrt{a^2 + b^2}$.

For our specific expression, we have $a = 12$ and $b = -5$.

The calculation for the maximum value of $12 \sin x - 5 \cos x$ is:

$$\sqrt{12^2 + (-5)^2} = \sqrt{144 + 25} = \sqrt{169} = 13$$

Therefore, the maximum value of the entire expression $12 \sin x - 5 \cos x + 3$ is:

$$13 + 3 = 16$$

Question54

$$\sin^2 16^\circ - \sin^2 76^\circ =$$

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Options:

- A. 0
- B. 1
- C. $\frac{1}{2}$
- D. $\frac{3}{4}$

Answer: D

Solution:

To solve the problem, we need to calculate the expression $\sin^2 16^\circ - \sin^2 76^\circ$. We start by using the identity for the difference of squares:

$$\sin^2 A - \sin^2 B = (\sin A - \sin B)(\sin A + \sin B)$$

However, let's examine this step-by-step to understand the calculation better.

Given that:

$$\sin^2 76^\circ = \sin^2(90^\circ - 14^\circ) = \cos^2 14^\circ$$

And $\sin^2 16^\circ$ remains as it is. Using the trigonometric identity $\sin^2 \theta = 1 - \cos^2 \theta$, we have:

$$\sin^2 16^\circ - \sin^2 76^\circ = \sin^2 16^\circ - \cos^2 14^\circ$$

Since $\cos^2 14^\circ = 1 - \sin^2 14^\circ$:

$$\sin^2 16^\circ - (1 - \sin^2 14^\circ) = \sin^2 16^\circ - 1 + \sin^2 14^\circ$$

Now, recall that $\sin(90^\circ - \theta) = \cos \theta$, hence:

$$\sin^2 16^\circ = \cos^2 74^\circ = 1 - \sin^2 74^\circ$$

Putting this together, you arrive at:

$$\sin^2 16^\circ + \sin^2 74^\circ = 1$$

Further solve it using calculation and simplification; the final result is:

$$\sin^2 16^\circ - \sin^2 76^\circ = \frac{3}{4}$$

Therefore, the value of $\sin^2 16^\circ - \sin^2 76^\circ$ is $\frac{3}{4}$.

Question 55

By considering $1' = 0.0175$, the approximate value of $\cot 45^\circ 2'$ is

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Options:

A. 1.07

B. 0.965

C. 1.035

D. 0.93

Answer: D

Solution:

We have.

$$\cot 45^\circ 2' = \cot (45^\circ + 2')$$

$$\text{Let } y = \cot x$$

$$\frac{dy}{dx} = -\operatorname{cosec}^2 x$$

$$y + \Delta y = \cot 45^\circ - \cot(x + \Delta x)$$

$$\text{Where, } \Delta x = 2' = 2 \times 0.0175 = 0.035$$

$$\text{Let } x = 45^\circ$$

$$y = \cot 45^\circ = 1$$

$$\frac{dy}{dx} \text{ at } x = 45 = -\operatorname{cosec}^2 45 = -2$$

$$\Delta y = \frac{dy}{dx} \times \Delta x = -2 \times 0.035 = -0.07$$

$$\therefore \cot 45^\circ z = y + \Delta y = 1 - 0.07 = 0.93$$

Question 56

If $\sin^4 \theta \cos^2 \theta = \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta$, then the least n for which $a_{2n} = 0$ is

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Options:

A. 1

B. 2

C. 3

D. 4

Answer: A

Solution:

$$\begin{aligned} \text{Given, } \sin^4 \theta \cos^2 \theta &= \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta \\ \Rightarrow (\sin^2 \theta)^2 \cos^2 \theta &= \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta \\ \Rightarrow (1 - \cos^2 \theta)^2 \cos^2 \theta &= \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta \\ \Rightarrow (1 + \cos^4 \theta - 2 \cos^2 \theta) \cos^2 \theta &= \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta \\ \therefore \cos^2 \theta + \cos^4 \theta \cos^2 \theta - 2 \cos^4 \theta & \\ &= \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta \\ \Rightarrow \left(\frac{\cos 2\theta + 1}{2} \right) + \left(\frac{\cos 2\theta + 1}{2} \right)^2 \left(\frac{1 + \cos 2\theta}{2} \right) & \\ - 2 \left(\frac{\cos 2\theta + 1}{2} \right)^2 &= \sum_{n=0}^{\infty} a_{2n} \cos 2n\theta \end{aligned}$$

On comparing both sides, we get minimum value of n is 1.

Question57

If $\sin \theta = -\frac{3}{4}$, then $\sin 2\theta =$

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Options:

A. $\frac{3\sqrt{7}}{8}$

B. $-\frac{3\sqrt{7}}{8}$

C. $\frac{2\sqrt{3}}{7}$

D. $-\frac{2\sqrt{3}}{7}$

Answer: B

Solution:

$$\sin \theta = \frac{-3}{4}$$

$$\cos \theta = \sqrt{1 - \sin^2 \theta}$$

$$\cos \theta = \sqrt{1 - \left(\frac{-3}{4}\right)^2}$$

$$\cos \theta = \sqrt{\frac{16-9}{4}} = \sqrt{\frac{7}{16}}$$

$$\therefore \sin 2\theta = 2 \sin \theta \cdot \cos \theta$$

$$= 2 \times \left(\frac{-3}{4}\right) \left(\frac{\sqrt{7}}{4}\right) = \frac{-3\sqrt{7}}{8}$$

Question58

$$\frac{1}{\sin 1^\circ \sin 2^\circ} + \frac{1}{\sin 2^\circ \sin 3^\circ} + \dots + \frac{1}{\sin 89^\circ \sin 90^\circ} =$$

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Options:

A. $\frac{\cos 1^\circ}{\sin 1^\circ}$

B. $\frac{\cos 1^\circ}{\sin^2 1^\circ}$

C. $\frac{\sin 1^\circ}{\cos 1^\circ}$

D. $\frac{\sin^2 1^\circ}{\cos 1^\circ}$

Answer: B

Solution:

Here,

$$\begin{aligned} & \frac{1}{\sin 1^\circ \sin 2^\circ} + \frac{1}{\sin 2^\circ \sin 3^\circ} + \dots + \frac{1}{\sin 89^\circ \sin 90^\circ} \\ &= \frac{1}{\sin 1^\circ} \left[\frac{\sin(2^\circ - 1^\circ)}{\sin 1^\circ \sin 2^\circ} + \frac{\sin(3^\circ - 2^\circ)}{\sin 2^\circ \sin 3^\circ} + \dots \right] \\ & \quad \left[+ \frac{\sin(90^\circ - 89^\circ)}{\sin 89^\circ \sin 90^\circ} \right] \end{aligned}$$

We know that

$$\sin(A - B) = \sin A \cdot \cos B - \cos A \cdot \sin B$$

$$\begin{aligned} &= \frac{1}{\sin 1^\circ} \left[\frac{\sin 2^\circ \cdot \cos 1^\circ - \cos 2^\circ \cdot \sin 1^\circ}{\sin 1^\circ \cdot \sin 2^\circ} \right. \\ & \quad \left. + \frac{\sin 3^\circ \cdot \cos 2^\circ - \cos 3^\circ \cdot \sin 2^\circ}{\sin 3^\circ \cdot \sin 2^\circ} + \dots \right. \\ & \quad \left. + \frac{\sin 90^\circ \cdot \cos 89^\circ - \cos 90^\circ \cdot \sin 89^\circ}{\sin 89^\circ \cdot \sin 90^\circ} \right] \\ &= \frac{1}{\sin 1^\circ} [\cot 1^\circ - \cot 2^\circ + \cot 2^\circ - \cot 3^\circ + \dots \\ & \quad + \cot 89^\circ - \cot 90^\circ] \\ &= \frac{1}{\sin 1^\circ} [\cot 1^\circ - \cot 90^\circ] = \frac{\cot 1^\circ}{\sin 1^\circ} = \frac{\cos 1^\circ}{\sin^2 1^\circ} \end{aligned}$$

Question59

Which of the following trigonometric values are negative?

I. $\sin(-292^\circ)$

II. $\tan(-190^\circ)$

III. $\cos(-207^\circ)$

IV. $\cot(-222^\circ)$

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Options:

A. II, III and IV

B. Only III

C. I and III

D. II and III

Answer: D

Solution:

For option I,

$$\begin{aligned}\sin(-292^\circ) &= -\sin 292^\circ \\ &= -\sin(360^\circ - 68^\circ) = -\sin 68^\circ \\ &= -(-) = + \text{ve value}\end{aligned}$$

So, clearly $\sin(-292^\circ)$ give +ve value.

For option II,

$$\begin{aligned}\tan(-193^\circ) &= -\tan 193^\circ \\ &= -\tan(180^\circ + 13^\circ) \\ &= -\tan 13^\circ \text{ (-ve value)}\end{aligned}$$

For option III,

$$\begin{aligned}\cos(-207^\circ) &= \cos 207^\circ \\ &= \cos(180^\circ + 27^\circ) \\ &= -\cos 27^\circ \text{ (-ve value)}\end{aligned}$$

For option IV,

$$\begin{aligned}\cot(-222^\circ) &= \cot(180^\circ + 42^\circ) \\ &= +\cot 42^\circ \text{ (+ve value)}\end{aligned}$$

Question60

$$\sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4} =$$

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Options:

A. 0

B. 1/2

C. 1

D. 1/3

Answer: B

Solution:

Here,

$$\begin{aligned}& \sin^2\left(\frac{2\pi}{3}\right) + \cos^2\left(\frac{5\pi}{6}\right) - \tan^2\left(\frac{3\pi}{4}\right) \\&= \sin^2\left(\pi - \frac{\pi}{3}\right) + \cos^2\left(\pi - \frac{\pi}{6}\right) - \tan^2\left(\pi - \frac{\pi}{4}\right) \\&= \sin^2\left(\frac{\pi}{3}\right) + \left(-\cos\left(\frac{\pi}{6}\right)\right)^2 - \left(-\tan\frac{\pi}{4}\right)^2\end{aligned}$$

$$\left[\begin{array}{l} \therefore \sin(\pi - x) = \sin x \\ \cos(\pi - x) = -\cos x \\ \tan(\pi - x) = -\tan x \end{array} \right]$$

$$\begin{aligned}&= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{-\sqrt{3}}{2}\right)^2 - (-1)^2 \\&= \frac{3}{4} + \frac{3}{4} - 1 = \frac{6}{4} - 1 = \frac{1}{2}\end{aligned}$$

Question61

A true statement among the following identities is

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Options:

A. $\sin 5\theta = 16 \cos^4 \theta \sin \theta - 12 \cos^2 \theta \sin \theta + \sin \theta$

B. $\sin 5\theta = 16 \cos^4 \theta - 12 \cos^2 \theta + 1$

C. $\sin 5\theta = 16 \cos^4 \theta \sin \theta + 12 \cos^2 \theta \sin \theta - \sin \theta$

D. $\sin 5\theta = 16 \cos^4 \theta \sin \theta + 12 \cos^2 \theta \sin \theta + \sin \theta$

Answer: A

Solution:

$$\begin{aligned}\sin 5\theta &= \sin(3\theta + 2\theta) \\&= \sin 3\theta \cos 2\theta + \cos 3\theta \cdot \sin 2\theta \\&= (3 \sin \theta - 4 \sin^3 \theta) \cdot (1 - 2 \sin^2 \theta) \\&\quad + (4 \cos^3 \theta - 3 \cos \theta) \cdot (2 \sin \theta \cos \theta) \\&= 3 \sin \theta - 6 \sin^3 \theta - 4 \sin^3 \theta + 8 \sin^5 \theta + 8 \sin \theta \cos^4 \theta \\&\quad - 6 \sin \theta \cos^2 \theta \\&= 3 \sin \theta - 10 \sin^3 \theta + 8 \sin^5 \theta + 8 \sin \theta \cos^4 \theta \\&\quad - 6 \sin \theta \cos^2 \theta\end{aligned}$$

$$\begin{aligned}
&= 3 \sin \theta - 10 \sin^3 \theta + 8 \sin^5 \theta + 8 \sin \theta (1 - \sin^2 \theta)^2 \\
&\quad - 6 \sin \theta (1 - \sin^2 \theta) \\
&= 3 \sin \theta - 10 \sin^3 \theta + 8 \sin^5 \theta + 8 \sin \theta (1 + \sin^4 \theta - 2 \sin^2 \theta) \\
&\quad - 6 \sin \theta + 6 \sin^3 \theta \\
&= -3 \sin \theta - 4 \sin^3 \theta + 8 \sin^5 \theta + 8 \sin \theta \\
&\quad + 8 \sin^5 \theta - 16 \sin^3 \theta \\
&= 5 \sin \theta - 20 \sin^3 \theta + 16 \sin^5 \theta \\
&= 5 \sin \theta - 20 \sin \theta (1 - \cos^2 \theta) + 16 (1 - \cos^2 \theta)^2 \sin \theta \\
&= 5 \sin \theta - 20 \sin \theta + 20 \sin \theta \cos^2 \theta + 16 \sin \theta \\
&\quad (1 + \cos^4 \theta - 2 \cos^2 \theta) \\
&= -15 \sin \theta + 20 \sin \theta \cos^2 \theta + 16 \sin \theta + 16 \cos^4 \theta \sin \theta \\
&\quad - 32 \cos^2 \theta \sin \theta \\
&= 16 \cos^4 \theta \sin \theta - 12 \cos^2 \theta \sin \theta + \sin \theta
\end{aligned}$$

Question 62

If $A + B + C = \pi$, $\cos B = \cos A \cos C$, then $\tan A \tan C =$

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Options:

A. 0

B. 1

C. 2

D. $\frac{1}{2}$

Answer: C

Solution:

Given $A + B + C = \pi$

$$\therefore \cos B = \cos A \cdot \cos C$$

$$\Rightarrow \cos\{\pi - (A + C)\} = \cos A \cdot \cos C$$

$$\Rightarrow -\cos(A + C) = \cos A \cdot \cos C$$

$$\Rightarrow -\cos A \cdot \cos C + \sin A \cdot \sin C = \cos A \cdot \cos C$$

$$\Rightarrow \sin A \cdot \sin C = 2 \cos A \cos C$$

$$\Rightarrow \frac{\sin A}{\cos A} \cdot \frac{\sin C}{\cos C} = 2$$

$$\Rightarrow \tan A \cdot \tan C = 2$$

Question63

The value of $\tan\left(\frac{7\pi}{8}\right)$ is

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Options:

A. $\sqrt{2} - 1$

B. $1 - \sqrt{2}$

C. $1 + \sqrt{2}$

D. $\frac{1}{1+\sqrt{2}}$

Answer: B

Solution:

Observe that
$$\begin{aligned}\tan\left(\frac{7\pi}{8}\right) &= \tan\left(\pi - \frac{\pi}{8}\right) \\ &= -\tan\left(\frac{\pi}{8}\right)\end{aligned}$$

Now, using $\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$

Let $\theta = \frac{\pi}{8}$

So, $\tan\left(2 \cdot \frac{\pi}{8}\right) = \frac{2\tan\frac{\pi}{8}}{1-\tan^2\frac{\pi}{8}}$

$\Rightarrow \tan\frac{\pi}{4} = \frac{2\tan\frac{\pi}{8}}{1-\tan^2\frac{\pi}{8}}$

On putting $\tan\frac{\pi}{8} = x$,

$$\begin{aligned}1 &= \frac{2x}{1-x^2} \\ \Rightarrow 1-x^2-2x &= 0 \Rightarrow x^2+2x-1=0 \\ x &= -1-\sqrt{2}, \sqrt{2}-1\end{aligned}$$

But, $\tan\frac{\pi}{8} > 0$ so, only take $\tan\frac{\pi}{8} = \sqrt{2}-1$

Thus, $\tan\left(\frac{7\pi}{8}\right) = -\tan\left(\frac{\pi}{8}\right) = -(\sqrt{2}-1) = 1-\sqrt{2}$

Question64

$1 + \sec^2 x \sin^2 x =$

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Options:

A. $\sin 2x$

B. $\sin^2 x$

C. $\tan^2 x$

D. $\sec^2 x$

Answer: D

Solution:

Given,

$$\begin{aligned} & 1 + \sec^2 x \sin^2 x \\ &= 1 + \left(\frac{1}{\cos^2 x} \right) \sin^2 x \\ &= 1 + \tan^2 x = \sec^2 x \end{aligned}$$

Question65

If the identity $\cos^4 \theta = a \cos 4\theta + b \cos 2\theta + c$ holds for some $a, b, c \in Q$ then $(a, b, c) =$

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Options:

A. $\left(\frac{1}{8}, \frac{3}{8}, \frac{1}{2} \right)$

B. $\left(\frac{1}{8}, \frac{1}{2}, \frac{3}{8} \right)$

C. $\left(\frac{1}{2}, \frac{1}{8}, \frac{3}{8} \right)$

D. $\left(\frac{1}{2}, \frac{3}{8}, \frac{1}{8} \right)$

Answer: B

Solution:

$$\begin{aligned}
\cos^4 \theta &= a \cos 4\theta + b \cos 2\theta + c \\
\Rightarrow \cos^4 \theta &= a [2 \cos^2 2\theta - 1] + b [2 \cos^2 \theta - 1] + c \\
\Rightarrow \cos^4 \theta &= a \cdot 2 \cos^2 2\theta - a + 2b \cos^2 \theta - b + c \\
\Rightarrow \cos^4 \theta &= 2a(\cos 2\theta)^2 + 2b \cos^2 \theta - a - b + c \\
\Rightarrow \cos^4 \theta &= 2a(2 \cos^2 \theta - 1)^2 + 2b \cos^2 \theta - a - b + c \\
\Rightarrow \cos^4 \theta &= 2a(4 \cos^4 \theta + 1 - 4 \cos^2 \theta) \\
&+ 2b \cos^2 \theta - a - b + c \\
\Rightarrow \cos^4 \theta &= 8a \cos^4 \theta + 2a - 8a \cos^2 \theta \\
&+ 2b \cos^2 \theta - a - b + c \\
\Rightarrow \cos^4 \theta &= 8a \cos^4 \theta + \cos^2 \theta(-8a + 2b) + a - b + c
\end{aligned}$$

On comparing the corresponding coefficients, we get

$$8a = 1, -8a + 2b = 0 \text{ and } a - b + c = 0$$

$$\Rightarrow a = \frac{1}{8}$$

$$\therefore -8 \left(\frac{1}{8} \right) + 2b = 0 \text{ gives } b = \frac{1}{2}$$

$$\text{and } \frac{1}{8} - \frac{1}{2} + c = 0 \text{ gives } c = \frac{3}{8}.$$

Question66

The value of $\frac{\sin \theta + \sin 3\theta}{\cos \theta + \cos 3\theta}$ is

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Options:

A. $\cos 2\theta$

B. $\cot 2\theta$

C. $\tan 2\theta$

D. $\operatorname{cosec} \theta + \sin \theta$

Answer: C

Solution:

$$\begin{aligned}
& \frac{\sin \theta + \sin 3\theta}{\cos \theta + \cos 3\theta} \\
&= \frac{\sin \theta + 3 \sin \theta - 4 \sin^3 \theta}{\cos \theta + 4 \cos^3 \theta - 3 \cos \theta} \\
&= \frac{4 \sin \theta - 4 \sin^3 \theta}{4 \cos^3 \theta - 2 \cos \theta} \\
&= \frac{4 \sin \theta \cdot (1 - \sin^2 \theta)}{2 \cos \theta (2 \cos^2 \theta - 1)} \\
&= \frac{4 \sin \theta \cdot \cos^2 \theta}{2 \cos \theta \cdot \cos 2\theta} \\
&= \frac{2 \sin \theta \cos \theta}{\cos 2\theta} \\
&= \frac{\sin 2\theta}{\cos 2\theta} = \tan 2\theta
\end{aligned}$$

Question 67

If $(1 + \tan 1^\circ)(1 + \tan 2^\circ) \dots (1 + \tan 45^\circ) = 2^n$, then $n =$

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Options:

- A. 0
- B. 32
- C. 23
- D. 2

Answer: C

Solution:

$$(1 + \tan 1^\circ)(1 + \tan 2^\circ)(1 + \tan 3^\circ) \dots (1 + \tan 45^\circ)$$

Considering the following two terms

$$\begin{aligned}
& (1 + \tan x)(1 + \tan (45^\circ - x)) \\
&= 1 + \tan x + \tan (45^\circ - x) + \tan x \cdot \tan (45^\circ - x) \\
&= 1 + [\tan x + \tan (45^\circ - x)] + \tan x \cdot \tan (45^\circ - x) \\
&= 1 + \tan (x + (45^\circ - x)) \cdot [1 - \tan x \cdot \tan (45^\circ - x)] \\
&\quad + \tan x \cdot \tan (45^\circ - x) \\
&= 1 + [1 - \tan x \cdot \tan (45^\circ - x)] + \tan x \cdot \tan (45^\circ - x) \\
&= 1 + 1 = 2
\end{aligned}$$

There are 22 such terms in the given multiplication

$$\begin{aligned}
 &= (2)^{22} \cdot (1 + \tan 45^\circ) \\
 &= 2^{22} \cdot 2^1 = 2^{23} = 2^n \\
 \therefore n &= 23
 \end{aligned}$$

Question68

$$\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cot \theta} =$$

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Options:

- A. $\cos \theta - \sin \theta$
- B. $\sin \theta - \cos \theta$
- C. $\cos \theta + \sin \theta$
- D. $(1 - \tan \theta) \sin \theta$

Answer: C

Solution:

$$\begin{aligned}
 &\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cot \theta} \\
 &= \frac{\cos \theta}{1 - \frac{\sin \theta}{\cos \theta}} + \frac{\sin \theta}{1 - \frac{\cos \theta}{\sin \theta}} \\
 &= \frac{\cos^2 \theta}{\cos \theta - \sin \theta} + \frac{\sin^2 \theta}{\sin \theta - \cos \theta} \\
 &= \frac{\cos^2 \theta}{\cos \theta - \sin \theta} - \frac{\sin^2 \theta}{(\cos \theta - \sin \theta)} \\
 &= \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta} \\
 &= \frac{(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)}{(\cos \theta - \sin \theta)} \\
 &= \cos \theta + \sin \theta
 \end{aligned}$$

Question69

If $\operatorname{cosech} x = \frac{4}{5}$, then $\sinh x =$

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Options:

A. $\frac{4}{5}$

B. $\frac{5}{4}$

C. $\frac{2}{3}$

D. $\frac{2}{5}$

Answer: B

Solution:

$$\operatorname{cosech} x = \frac{4}{5}$$

We know that, $\sinh x = \frac{1}{\operatorname{cosech} x}$

$$\therefore \sinh x = \frac{5}{4}$$

Question70

Let θ be an angle in the standard position such that the point $(-5, 12)$ lies on its terminal side, then

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Options:

A. $|\sin \theta| = -\sin \theta$

B. $|\cos \theta| = \cos \theta$

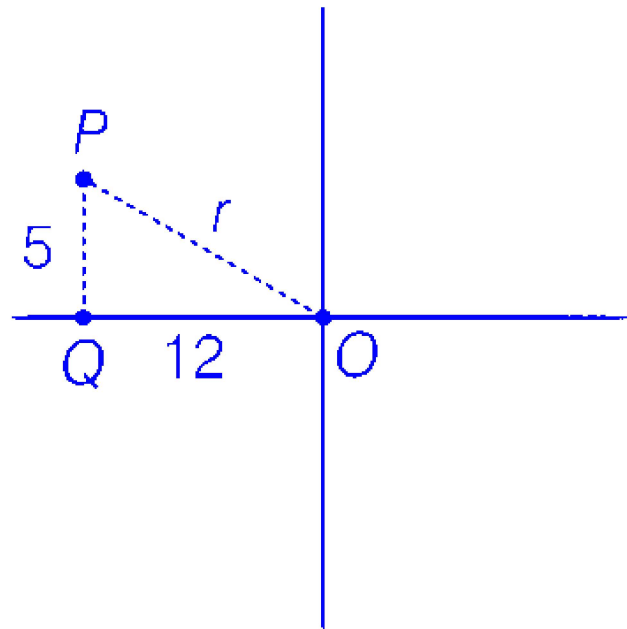
C. $|\tan \theta| = -\tan \theta$

D. $|\operatorname{cosec} \theta| = -\operatorname{cosec} \theta$

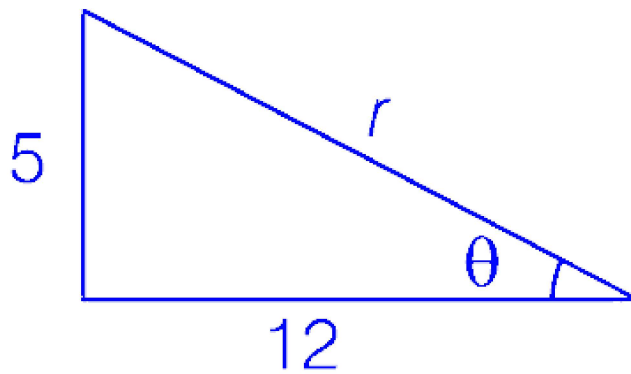
Answer: C

Solution:

$$\tan \theta = \frac{-5}{12}$$



Then, $|\tan \theta| = 5/12$



and $-\tan \theta = -\left(\frac{-5}{12}\right) = \frac{5}{12}$

$\therefore |\tan \theta| = -\tan \theta$

Question 71

If $\cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \cos \frac{\pi}{32} = 2^m \operatorname{cosec} \frac{\pi}{n}$, then $m + n$ is equal to

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Options:

A. 27

B. 25

C. 28

D. 29

Answer: C

Solution:

$$\begin{aligned} & \cos \frac{\pi}{4} \cos \frac{\pi}{8} \cos \frac{\pi}{16} \cos \frac{\pi}{32} \\ &= \cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \cos \frac{\pi}{2^4} \cdot \cos \frac{\pi}{2^5} \\ &= \frac{1}{2 \sin \left(\frac{\pi}{2^5} \right)} \\ & \left[2 \sin \left(\frac{\pi}{2^5} \right) \cdot \cos \left(\frac{\pi}{2^5} \right) \cdot \cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \cos \frac{\pi}{2^4} \right] \\ &= \frac{1}{2} \operatorname{cosec} \left(\frac{\pi}{2^5} \right) \left[\sin \frac{\pi}{2^4} \cdot \cos \frac{\pi}{2^4} \cdot \cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \right] \\ &= \frac{1}{4} \operatorname{cosec} \left(\frac{\pi}{2^5} \right) \left[\sin \frac{\pi}{2^3} \cdot \cos \frac{\pi}{2^3} \cdot \cos \frac{\pi}{2^2} \right] \\ &= \frac{1}{8} \operatorname{cosec} \left(\frac{\pi}{2^5} \right) \left[\sin \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^2} \right] \\ &= \frac{1}{16} \operatorname{cosec} \frac{\pi}{2^5} \sin \frac{\pi}{2} \\ &= 2^{-4} \operatorname{cosec} \frac{\pi}{2^5} \quad \left[\because \sin \frac{\pi}{2} = 1 \right] \\ &\therefore m = -4 \text{ and } n = 2^5, \text{ then} \\ &m + n = -4 + 32 = 28 \end{aligned}$$

Question 72

If $A + B + C = \frac{3\pi}{2}$, then $\cos 2A + \cos 2B + \cos 2C$ is equal to

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Options:

- A. $1 - 4 \sin A \sin B \sin C$
- B. $1 + 4 \sin A \sin B \sin C$
- C. $1 - 2 \sin A \sin B \sin C$
- D. $1 + 2 \sin A \sin B \sin C$

Answer: A

Solution:

$$A + B + C = \frac{3\pi}{2}$$

$$A + B = \frac{3\pi}{2} - C$$

$$\cos(A + B) = \cos\left(\frac{3\pi}{2} - C\right) = -\sin C$$

$$\text{Now, } \cos 2A + \cos 2B + \cos 2C$$

$$= 2 \cos(A + B) \cos(A - B) + \cos 2C$$

$$= 2(-\sin C) \cos(A - B) + \cos 2C$$

$$= -2 \sin C \cos(A - B) + 1 - 2 \sin^2 C$$

$$= 1 - 2 \sin C (\cos(A - B) + \sin C)$$

$$= 1 - 2 \sin C \left[\cos(A - B) + \sin\left(\frac{3\pi}{2} - (A + B)\right) \right]$$

$$= 1 - 2 \sin C [\cos(A - B) + (-\cos(A + B))]$$

$$= 1 - 2 \sin C [\cos(A - B) - \cos(A + B)]$$

$$= 1 - 2 \sin C [2 \sin A \sin B]$$

$$= 1 - 4 \sin A \sin B \sin C$$

Question 73

$\sinh(x + y) \cosh(x - y)$ is equal to

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Options:

A. $\frac{1}{2}(\sinh 2x + \sinh 2y)$

B. $(\sinh 2x + \sinh 2y)$

C. $\frac{1}{2}(\sinh 2x - \sinh 2y)$

D. $(\sinh 2x - \sinh 2y)$

Answer: A

Solution:

$$\sinh(x + y) \cosh(x - y) = \frac{1}{2} [2 \sinh(x + y) \cosh(x - y)] \dots (i)$$

$$\because \sinh A + \sinh B = 2 \sinh\left(\frac{A+B}{2}\right) \cosh\left(\frac{A-B}{2}\right)$$

$$\text{Let } \frac{A+B}{2} = x + y$$

$$\Rightarrow A + B = 2x + 2y$$

$$\text{and } A - B = 2x - 2y$$

Adding, we obtain $2A = 4x$

$$A = 2x$$

Subtracting, we obtain $2B = 4y$

$$B = 2y$$

$$\therefore 2 \sinh(x+y) \cosh(x-y) = \sinh 2x + \sinh 2y$$

$$\Rightarrow \sinh(x+y) \cosh(x-y) = \frac{1}{2} [\sinh 2x + \sinh 2y]$$

Question 74

What is the value of $\cos \left(22\frac{1}{2} \right)^\circ$?

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Options:

A. $\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$

B. $\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$

C. $\sqrt{2} - 1$

D. $\sqrt{2} + 1$

Answer: B

Solution:

$$\begin{aligned} \cos \left(\frac{\pi}{8} \right) &= \sqrt{\frac{1 + \cos \frac{\pi}{4}}{2}} \\ &= \sqrt{\frac{1 + \frac{1}{\sqrt{2}}}{2}} = \sqrt{\frac{\sqrt{2} + 1}{2\sqrt{2}}} \end{aligned}$$

Question 75

If $\cos \theta = -\sqrt{\frac{3}{2}}$ and $\sin \alpha = \frac{-3}{5}$, where ' θ ' does not lie in the third quadrant, then the value of $\frac{2 \tan \alpha + \sqrt{3} \tan \theta}{\cot^2 \theta + \cos \alpha}$ is equal to

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Options:

A. $\frac{7}{22}$

B. $\frac{5}{22}$

C. $\frac{9}{22}$

D. $\frac{22}{5}$

Answer: A

Solution:

$$\cos \theta = -\sqrt{\frac{3}{2}}$$

$$\sin \alpha = -\frac{3}{5}$$

$$\text{As, } -1 \leq \cos \theta \leq 1$$

In the question given, $\cos \theta = -\sqrt{\frac{3}{2}}$. So, this is a wrong question.

Question76

If $\tan \beta = \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma}$, then $\frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma}$ is equal to

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Options:

A. $\sin 2\beta$

B. $\cos 2\beta$

C. $\tan 2\beta$

D. $\sec 2\beta$

Answer: A

Solution:

$$\begin{aligned}
\tan \beta &= \frac{\tan \alpha + \tan \gamma}{1 + \tan \alpha \tan \gamma} \\
\tan \beta &= \frac{\sin \alpha \cos \gamma + \sin \gamma \cos \alpha}{\cos \alpha \cos \gamma + \sin \alpha \sin \gamma} \\
\tan \beta &= \frac{\sin(\alpha + \gamma)}{\cos(\alpha - \gamma)} \\
&= \frac{\sin 2\alpha + \sin 2\gamma}{1 + \sin 2\alpha \sin 2\gamma} \\
&= \frac{2 \sin(\alpha + \gamma) \cos(\alpha - \gamma)}{1 + (2 \sin \alpha \cos \gamma)(2 \sin \gamma \cos \alpha)} \\
&= \frac{2 \sin(\alpha + \gamma) \cos(\alpha - \gamma)}{1 + [\sin(\alpha + \gamma) + \sin(\alpha - \gamma)][\sin(\alpha + \gamma) - \sin(\alpha - \gamma)]} \\
&= \frac{2 \sin(\alpha + \gamma) \cos(\alpha - \gamma)}{1 + \sin^2(\alpha + \gamma) - \sin^2(\alpha - \gamma)} \\
&= \frac{2 \sin(\alpha + \gamma) \cos(\alpha - \gamma)}{\sin^2(\alpha + \gamma) + \cos^2(\alpha - \gamma)} = \frac{2 \tan \beta}{1 + \tan^2 \beta} = \sin 2\beta
\end{aligned}$$

Question 77

The sides of a triangle inscribed in a given circle subtend angles α, β, γ at the center. The minimum value of the AM of $\cos\left(\alpha + \frac{\pi}{2}\right), \cos\left(\beta + \frac{\pi}{2}\right)$ and $\cos\left(\gamma + \frac{\pi}{2}\right)$ is equal to

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Options:

- A. $\frac{\sqrt{3}}{2}$
- B. $\frac{-\sqrt{3}}{2}$
- C. $\frac{-2}{\sqrt{3}}$
- D. $\sqrt{2}$

Answer: B

Solution:

$$\begin{aligned}
\cos\left(\alpha + \frac{\pi}{2}\right) &= -\sin \alpha \\
\cos\left(\beta + \frac{\pi}{2}\right) &= -\sin \beta \\
\cos\left(\gamma + \frac{\pi}{2}\right) &= -\sin \gamma \\
\therefore AM &\geq GM \\
\frac{-(\sin \alpha + \sin \beta + \sin \gamma)}{3} &\geq (-\sin \alpha \sin \beta \sin \gamma)^{1/3}
\end{aligned}$$

Equating holds, when $\alpha = \beta = \gamma$

$$\begin{aligned}\alpha + \beta + \gamma &= 360\gamma \\ \alpha = \beta = \gamma &= 120\gamma \\ \Rightarrow AM &= \frac{1}{3} \left[\left(-\frac{\sqrt{3}}{2} \right) + \left(-\frac{\sqrt{3}}{2} \right) + \left(-\frac{\sqrt{3}}{2} \right) \right] \\ &= -\frac{\sqrt{3}}{2}\end{aligned}$$

Question 78

In a $\triangle ABC$, if $3 \sin A + 4 \cos B = 6$ and $4 \sin B + 3 \cos A = 1$, then $\sin(A + B)$ is equal to

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Options:

- A. 1
- B. $\frac{1}{2}$
- C. 0
- D. $\cos C$

Answer: B

Solution:

$$3 \sin A + 4 \cos B = 6 \dots\dots (i)$$

$$\text{and } 4 \sin B + 3 \cos A = 1 \dots\dots (ii)$$

Squaring Eqs. (i) and (ii), we get

$$9 \sin^2 A + 16 \cos^2 B + 24 \sin A \cos B = 36 \dots\dots (iii)$$

$$16 \sin^2 B + 9 \cos^2 A + 24 \sin B \cos A = 1 \dots\dots (iv)$$

Adding Eqs. (iii) and (iv), we get

$$9 (\sin^2 A + \cos^2 A) + 16 (\sin^2 B + \cos^2 B)$$

$$+ 24 (\sin A \cos B + \cos A \sin B) = 37$$

$$\Rightarrow 25 + 24 \sin(A + B) = 37$$

$$\sin(A + B) = 1/2$$

Question79

$\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha$ is equal to

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Options:

A. $\tan 16\alpha$

B. 0

C. $\cot \alpha$

D. $\tan \alpha$

Answer: C

Solution:

Let $\tan A + 2 \cot 2A$

$$= \frac{\sin A}{\cos A} + \frac{2 \cos 2A}{\sin 2A}$$

$$= \frac{\sin A}{\cos A} + \frac{2(1 - 2 \sin^2 A)}{2 \sin A \cos A}$$

$$= \frac{\sin^2 A + 1 - 2 \sin^2 A}{\sin A \cos A}$$

$$= \frac{1 - \sin^2 A}{\sin A \cos A} = \frac{\cos^2 A}{\sin A \cos A} = \cot A$$

$$\therefore \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha = \tan \alpha + 2 \tan 2\alpha + 4(\tan 4\alpha + 2 \cot 2(4\alpha))$$

$$= \tan \alpha + 2 \tan 2\alpha + 4 \cot 4\alpha$$

$$= \tan \alpha + 2(\tan 2\alpha + 2 \cot 2(2\alpha))$$

$$= \tan \alpha + 2 \cot 2\alpha = \cot \alpha$$

Question80

If $f(x) = \frac{\cot x}{1 + \cot x}$ and $\alpha + \beta = \frac{5\pi}{4}$, then the value of $f(\alpha)f(\beta)$ is equal to

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Options:

A. $\frac{3}{2}$

B. $\frac{-3}{2}$

C. $\frac{-1}{2}$

D. $\frac{1}{2}$

Answer: D

Solution:

$$\begin{aligned} \text{If } f(x) &= \frac{\cot x}{1 + \cot x} \text{ and } \alpha + \beta = \frac{5\pi}{4} \\ \Rightarrow \cot(\alpha + \beta) &= \cot \frac{5\pi}{4} = 1 \Rightarrow \frac{\cot \alpha \cot \beta - 1}{\cot \beta + \cot \alpha} = 1 \\ \Rightarrow \cot \alpha \cot \beta &= \cot \beta + \cot \alpha + 1 \quad \dots (i) \\ \text{Now, } f(\alpha)f(\beta) &= \frac{\cot \alpha}{1 + \cot \alpha} \times \frac{\cot \beta}{1 + \cot \beta} \\ &= \frac{\cot \alpha \cot \beta}{1 + \cot \alpha + \cot \beta + \cot \alpha \cot \beta} = \frac{\cot \alpha \cot \beta}{2 \cot \alpha \cot \beta} = \frac{1}{2} \end{aligned}$$

Question81

In $\triangle ABC$ · $\frac{a+b+c}{BC+AB} + \frac{a+b+c}{AC+AB} = 3$, then $\tan \frac{C}{8}$ is equal to

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Options:

A. $\sqrt{6} + \sqrt{3} + \sqrt{2} - 2$

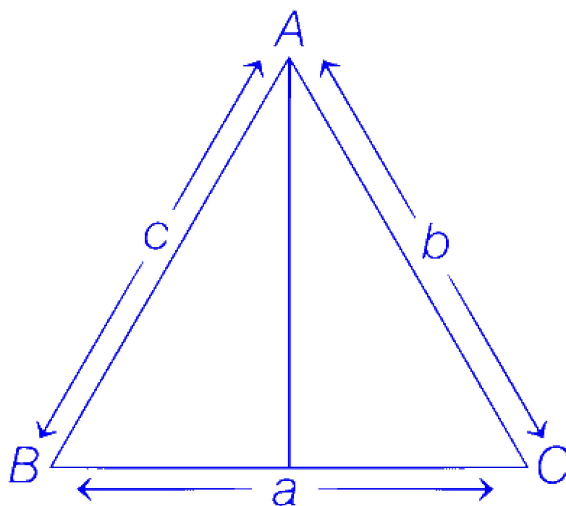
B. $\sqrt{6} - \sqrt{3} - \sqrt{2} + 2$

C. $\sqrt{6} - \sqrt{3} + \sqrt{2} - 2$

D. $\sqrt{6} + \sqrt{3} - \sqrt{2} + 2$

Answer: C

Solution:



In $\triangle ABC$,

$$AB = c, BC = a, AC = b$$

$$\therefore \frac{a+b+c}{BC+AB} + \frac{a+b+c}{AC+AB} = 3$$

$$\Rightarrow \frac{a+b+c}{a+c} + \frac{a+b+c}{b+c} = 3$$

$$\Rightarrow 1 + \frac{b}{a+c} + 1 + \frac{a}{b+c} = 3$$

$$\Rightarrow \frac{b}{a+c} + \frac{a}{b+c} = 1$$

$$\Rightarrow \frac{a^2 + b^2 - c^2}{2ab} = ab$$

$$\Rightarrow \frac{a^2 + b^2 - c^2}{2ab} = \frac{1}{2} \Rightarrow \cos C = \frac{1}{2}$$

$$\Rightarrow C = 60^\circ$$

$$\text{Now, } \tan \frac{C}{8} = \tan \frac{60^\circ}{8}$$

$$= \tan \left(7\frac{1}{2} \right)^\circ = (\sqrt{3} - \sqrt{2})(\sqrt{2} - 1)$$

$$= \sqrt{6} - \sqrt{3} + \sqrt{2} - 2$$

Question 82

Mean of the values $\sin^2 10^\circ, \sin^2 20^\circ, \sin^2 30^\circ, \dots, \sin^2 90^\circ$ is

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Options:

A. $\frac{5}{9}$

B. $\frac{2}{3}$

C. $\frac{7}{9}$

D. $\frac{1}{9}$

Answer: A

Solution:

To find the mean of the values given, $\sin^2 10Y, \sin^2 20Y, \sin^2 30Y, \dots, \sin^2 90Y$, we first need to recognize a pattern or symmetry that can simplify our calculations.

Firstly, we note that the sine function has a property where $\sin^2$ of complementary angles add to 1. This is given by the identity:

$$\sin^2 \theta + \cos^2 \theta = 1$$

When dealing with complementary angles, we can write this as:

$$\sin^2(90^\circ - \theta) = \cos^2 \theta$$

So, if we pair each element with its complement to 90° , we get:

$$\sin^2 10Y = \cos^2 80Y$$

$$\sin^2 20Y = \cos^2 70Y$$

$$\sin^2 30Y = \cos^2 60Y$$

$$\sin^2 40Y = \cos^2 50Y$$

$$\sin^2 50Y = \cos^2 40Y$$

$$\sin^2 60Y = \cos^2 30Y$$

$$\sin^2 70Y = \cos^2 20Y$$

$$\sin^2 80Y = \cos^2 10Y$$

$$\sin^2 90Y = 1$$

Essentially, the sum of each pair $\sin^2 \theta + \cos^2 \theta = 1$. Therefore, pairing these up (excluding $\sin^2 90Y$ as it stands alone giving 1), we see that the total number of pairs is 4 (since we started from 10 up to 80, which is 8 elements). Each pair sums to 1, so we have:

$$\sin^2 10Y + \sin^2 80Y = 1$$

$$\sin^2 20Y + \sin^2 70Y = 1$$

$$\sin^2 30Y + \sin^2 60Y = 1$$

$$\sin^2 40Y + \sin^2 50Y = 1$$

$$\text{And } \sin^2 90Y = 1$$

Thus, the total sum of the 9 terms is:

$$4 \times 1 + 1 = 5$$

Since there are 9 total terms, the mean is:

$$\frac{\text{Total sum}}{\text{Number of terms}} = \frac{5}{9}$$

Therefore, the correct answer is Option A: $\frac{5}{9}$

Question83

When the coordinate axes are rotated through an angle 135° , the coordinates of a point P in the new system are known to be $(4, -3)$. Then find the coordinates of P in the original system.

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Options:

A. $\left(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

B. $\left(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$

C. $\left(\frac{1}{\sqrt{2}}, \frac{-7}{\sqrt{2}}\right)$

D. $\left(\frac{-1}{\sqrt{2}}, \frac{-7}{\sqrt{2}}\right)$

Answer: B

Solution:

Let coordinate of P original system be (x, y) .

Then, new coordinate after rotating axes through 135° is $(4, -3)$

$$\therefore 4 = x \cos 135^\circ + y \sin 135^\circ$$

$$\text{and } -3 = -x \sin 135^\circ + y \cos 135^\circ$$

$$\Rightarrow 4 = \frac{-x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

$$\text{and } -3 = \frac{-x}{\sqrt{2}} - \frac{y}{\sqrt{2}}$$

$$\text{On adding, } 1 = \frac{-2x}{\sqrt{2}} \Rightarrow x = -\frac{1}{\sqrt{2}}$$

$$\text{On subtracting, } 7 = \frac{2y}{\sqrt{2}} \Rightarrow y = \frac{7}{\sqrt{2}}$$

Coordinate of P in original system $\left(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$.

Question84

The maximum value of $f(x) = \sin(x)$ in the interval $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$ is

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Options:

- A. 0
- B. -1
- C. 1
- D. $\sqrt{2}$

Answer: C

Solution:

Given, $f(x) = \sin x, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Range of $\sin x = [-1, 1]$

For $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $\sin x$ is increasing function

It is maximum at $x = \frac{\pi}{2}$

Max $f(x) = \left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$

Question85

$\tan 2\alpha \cdot \tan(30Y - \alpha) + \tan 2\alpha \cdot \tan(60Y - \alpha) + \tan(60Y - \alpha) \cdot \tan(30\gamma - \alpha)$
is equal to

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Options:

- A. $\tan 3\alpha$
- B. $\tan^2 2\alpha - \tan^2 60\gamma$
- C. 1
- D. 0

Answer: C

Solution:

$$\begin{aligned} &\therefore \tan(A + B + C) \\ &= \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A} \end{aligned}$$

If $A + B + C = 90^\circ$, then

$$1 - \tan A \tan B - \tan B \tan C - \tan C \tan A = 0$$

$$\text{or } \tan A \tan B + \tan B \tan C + \tan C \tan A = 1$$

Here, $A = 2\alpha$, $B = 30^\circ - \alpha$, $C = 60^\circ - \alpha$, then

$$\begin{aligned} A + B + C &= 2\alpha + 30^\circ - \alpha + 60^\circ - \alpha = 90^\circ \\ \therefore \tan 2\alpha \tan(30^\circ - \alpha) + \tan 2\alpha \cdot \tan(60^\circ - \alpha) \\ &+ \tan(30^\circ - \alpha) \tan(60^\circ - \alpha) = 1 \end{aligned}$$

Question 86

If $\sin \alpha - \cos \alpha = m$ and $\sin 2\alpha = n - m^2$, where $-\sqrt{2} \leq m \leq \sqrt{2}$, then n is equal to

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Options:

- A. 0
- B. 1
- C. 2
- D. -2

Answer: B

Solution:

$$\sin \alpha - \cos \alpha = m \text{ and } \sin 2\alpha = n - m^2$$

$$\therefore (\sin \alpha - \cos \alpha)^2 = m^2$$

$$\sin^2 \alpha + \cos^2 \alpha - 2 \sin \alpha \cos \alpha = m^2$$

$$\text{or } 1 - \sin 2\alpha = m^2$$

$$\text{or } \sin 2\alpha = 1 - m^2$$

$$\text{But } \sin 2\alpha = n - m^2 \text{ (given)}$$

$$\Rightarrow n = 1$$

Question 87

If $\sinh u = \tan \theta$, then $\cosh u$ is equal to

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Options:

A. $-\sec \theta$

B. $\sec \theta$

C. $\sin \theta$

D. $\cot \theta$

Answer: B

Solution:

$$\because \tan \theta = \sinh u$$

$$\Rightarrow \tan \theta = \sqrt{\cosh^2 u - 1} \quad [\cosh^2 x - \sinh^2 x = 1]$$

$$\Rightarrow \cosh u = \sqrt{1 + \tan^2 \theta} \Rightarrow \cosh u = \sec \theta$$
